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Favorable Displacement Conditions for Efficient Vertical Equilibrium Simulation of Gas Storage and Disposal in Aquifers

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Abstract

Vertical equilibrium (VE) simulation has received increasing attention in recent years as a fast and reliable approach for modeling CO₂ storage in aquifers; the approach is also applicable to other fluid systems under appropriate conditions. This study provides a unified theoretical framework for VE simulation to quantitatively assess the conditions under which the approach remains accurate. Both inspectional and numerical analyses are employed to achieve this end. The findings are summarized using simple closed-form expressions, dimensionless numbers, and regime maps to provide generalizable guidance for practical implementation. Application cases are briefly discussed at the end of the paper.

Introduction

Mitigating the anthropogenic temperature rise below 1.5-2 centigrade is the Paris Agreement goal for which IPCC has identified several solutions to reduce the anthropogenic greenhouse gases emission in the atmosphere (IPCC 2022). Geological CO₂ storage (GCS), or sequestration, is one of the promising solutions with demonstrated success on the megatonne-level and is projected to be on the gigatonne level to meet the objectives for which it is deployed (Zhang, Jackson, and Krevor 2022). GCS has been around since 1995 as an important measure targeting in the first place difficult to decarbonize industries (Watson et al. 1996). The basic technology model relies primarily on the concept of a geological storage complex where CO₂ can be injected, immobilized by hydrodynamic, residual, dissolution, and mineral trapping, and should be definitely contained safely. Among the various geological storage options, saline aquifers have the largest capacity to sequester CO₂, and they are the main focus of this study. In the context of energy transition, hydrogen storage in geological reservoirs is expected to play a crucial role in stabilizing fluctuating energy demand. Readers are referred to a recent review by Krevor et al. (2023) and the references therein. The present study deals with the hydrodynamics of two-phase flows, applicable to CO₂ and hydrogen storage in aquifers.

Supercritical CO₂ flow under typical deep saline aquifers conditions is prominently governed by gravity and capillarity forces (Guo et al. 2016; Taku Ide, Jessen, and Orr 2007), and observations have emphasized

the role of gravity in the long-term migration (Cavanagh 2013; Cowton et al. 2018; Eiken et al. 2011), but to a less extent during injection, see for example Grude et al. (2014). Simulation models are the prime means for assessing and developing a geological CO₂ storage project in such formations (Ajayi, Gomes, and Bera 2019). However, given the large density contrast, and the need to an accurate prediction means of the plume migration and pressure distribution, the regular coarse-grid, dispersed flow 3D simulators fail to perform at the accuracy required, mainly due to their sensitivity to the grid size, primarily vertically (Ligaarden and Nilsen 2010). This has been long observed in oil and gas reservoirs simulation (Aziz and Settari 1979). From here, the utilization of vertical equilibrium models emerged as a fast and reliable approach, should the conditions be in favour, to simulate CO₂ injection into aquifers, and possibly any other pair of fluids in subsurface formations with different physics (Alamara, Blondeau, and Bogdanov 2023c; Andersen, Nilsen, and Gasda 2017; Becker et al. 2017; Gasda, Nordbotten, and Celia 2009; Nilsen et al. 2011, 2017; Nilsen, Lie, and Andersen 2016). Vertical equilibrium simulation relies on the concept of vertically-integrating the flow equations and embedding analytically the flow dynamics and heterogeneity while reducing the model's grid sensitivity in the vertical direction to its minimum. It has been suggested initially as a way to reduce computational costs (Coats et al. 1967; Coats, Dempsey, and Henderson 1971). However, the way of its development is rich in different variants and, different researchers have used different terms and justifications, resulting in a plethora of specific cases theories and solutions (Ekraan 1992; Yortsos 1995).

Fig. 1 shows a comparison of exact same models (one is realistic, and the other is hypothetical) yet with different vertical grid resolutions. Obviously, the impact of grid resolution in the vertical direction can mislead the decision-making and, at the extreme case, fail to predict accurate setups to meet the lease regulations of not exceeding a certain area if sufficiently refined grids are not used. The key problem relates to the high computational power required by very fine three-dimensional (3D) models and inaccurate results obtained by coarse 3D ones. Vertical equilibrium simulation is basically an approach to completely eliminate the vertical discretization. The standing issue is to identify a working range of the validity and applicability of the approach. Given the complex spatial and temporal dependent interactions and dynamics between the fluids and the rocks, and the flow regimes involved in transporting CO₂, establishing quantitatively such a range cannot be made without many simplifying assumptions. Several attempts were provided in the literature, c.f. Coats et al. (1967); Coats et al. (1971); Nordbotten and Celia (2006); Nordbotten and Dahle (2011); Yortsos (1995) and qualitatively, Martin (1968). More recently, Vahabzadeh et al. (2025) investigated the applicability of VE models for hydrogen storage in aquifers and reported results based on dimensionless groups. In this study, we examine the various aspects of the vertical equilibrium (VE) simulation approach with the aim of establishing a general framework for its application and delineating the conditions under which it can be reliably employed. We begin by introducing the fundamental concepts and terminology, then proceed to numerical experiments and analyses. Finally, we briefly present two application cases: one based on real field data and another on a synthetic model.

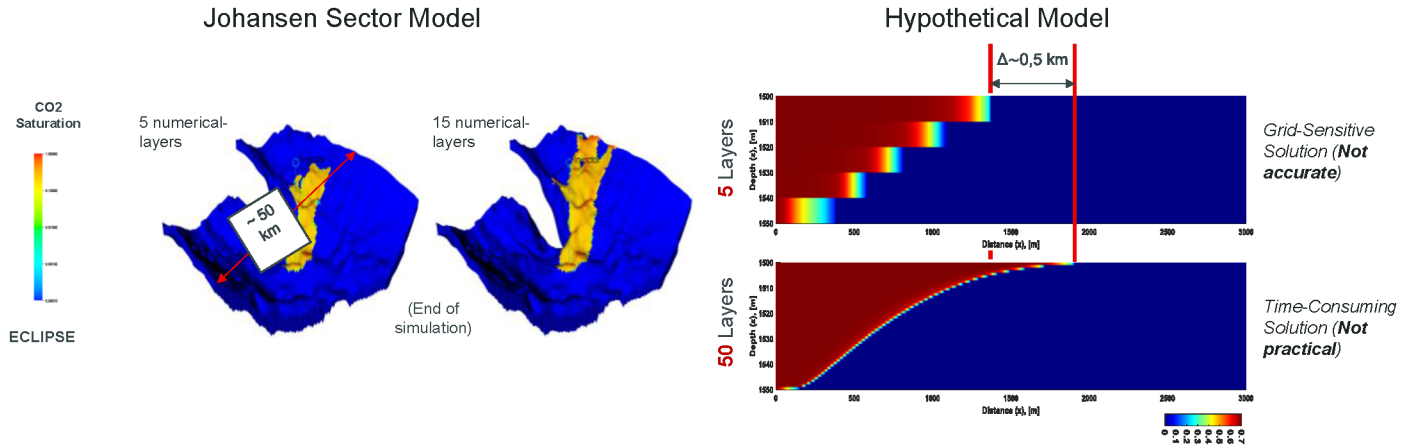


Figure 1—Impact of vertical grid sensitivity: (left) Johansen sector model, (right) Hypothetical model. Adapted from Alamara, Blondeau, Thibeau, et al. (2025)

Unified Theoretical Framework

In this section, we provide a concise formulation of the fundamental flow equations governing this study in dimensionless form. Then, we discuss the topic of dimensionality reduction and subsequently the flow regimes. The component-wise, differential mass conservation equation of phase j in cartesian coordinates (x, y, z) is given by:

$$\frac{\partial}{\partial t}(\phi \rho_j s_j) + \frac{\partial}{\partial x}(\rho_j u_j) + \frac{\partial}{\partial y}(\rho_j w_j) + \frac{\partial}{\partial z}(\rho_j v_j) = \rho_j q_{sj} \quad (1)$$

where ϕ is the porosity [$L^3 L^{-3}$], ρ is the mass density [ML^{-3}], s is the phase saturation [$L^3 L^{-3}$], u, w, v are the flux components [Lt^{-1}] in the x, y, z directions respectively, and q_s is the external mass flux [$L^3 L^{-3} t^{-1}$]. Eq. 1 is further facilitated for the purpose of the current discussion by introducing the phase isothermal compressibility $c_j = \rho_j^{-1} \partial \rho_j / \partial p_j|_T$ [$LM^{-1}t^2$] and the pore compressibility $c_\phi = \phi^{-1} \partial \phi / \partial p_j|_T$ where p_j is the phase pressure. Expanding the derivatives, using the compressibility definition, and rearranging gives:

$$\underbrace{\phi \frac{\partial s_j}{\partial t} + \frac{\partial u_j}{\partial x} + \frac{\partial w_j}{\partial y} + \frac{\partial v_j}{\partial z}}_{\text{Incompressible Flux Continuity}} + \underbrace{s_j \phi c_t \frac{\partial p_j}{\partial t} + c_j \left(u_j \frac{\partial p_j}{\partial x} + w_j \frac{\partial p_j}{\partial y} + v_j \frac{\partial p_j}{\partial z} \right)}_{\text{Pressure Variations}} = q_{sj} \quad (2)$$

where $c_t = c_j + c_\phi$ [$LM^{-1}t^2$] is the total compressibility. We scale the equation by introducing the following dimensionless variables:

$$t_D = q_t t / \phi s_{jm} L_x, \quad s_{jD} = (s_j - s_{jm}) / (s_{jM} - s_{jm}), \quad x_D = x / L_x, \quad y_D = y / L_y, \quad z_D = z / H, \quad p_{jD} = p_j k_h / q_t \mu_1 L_x \quad (3)$$

where q_t is a characteristic total flux, L_x is the characteristics lateral distance, H is the characteristic true thickness, s_{jm} and s_{jM} are the immobile and maximum mobile saturations respectively, determined by the phase and process of whether drainage or imbibition, k_h is the lateral permeability, and μ_1 is the resident phase's dynamic viscosity [$ML^{-1}t^{-1}$]. The dimensionless form is then:

$$\frac{\partial s_{jD}}{\partial t_D} + \frac{\partial u_{jD}}{\partial x_D} + \epsilon_1^{-1} \frac{\partial w_{jD}}{\partial y_D} + \epsilon_1^{-1} \frac{\partial v_{jD}}{\partial z_D} + N_{cm} \left[s_{jD} \frac{\partial p_{jD}}{\partial t_D} + c_{jD} \left(u_{jD} \frac{\partial p_{jD}}{\partial x_D} + w_{jD} \epsilon_1^{-1} \frac{\partial p_{jD}}{\partial y_D} + v_{jD} \epsilon_1^{-1} \frac{\partial p_{jD}}{\partial z_D} \right) \right] = q_{sjD} \quad (4)$$

Eq. 4 defines some relevant dimensionless numbers: (1) Two aspect ratios $\epsilon_1 = L_y / L_x$ and $\epsilon_2 = H / L_x$, (2) dimensionless compressibility number $N_{cm} = q_t \mu_1 L_x c_t / k_h$, (3) phase compressibility ratio $c_{jD} = c_j / c_t$, and (4) a dimensionless mass transfer number $q_{sjD} = q_{sj} L_x / q_t$. For $N_{cm} \rightarrow 0$, the explicit pressure dependency is removed from the equation. Furthermore, we focus on models where $\epsilon \ll 1$, or in other words, two-dimensional ($x-z$) slice models with no external mass flux other than boundary conditions representing wells, hence:

$$\frac{\partial s_{jD}}{\partial t_D} + \frac{\partial u_{jD}}{\partial x_D} + \epsilon_{\perp}^{-1} \frac{\partial v_{jD}}{\partial z_D} = 0. \quad (5)$$

The last expression implies:

$$\frac{\partial}{\partial x_D} \left(u_{tD} = \sum_{j=w,c} u_{jD} \right) + \epsilon_{\perp}^{-1} \frac{\partial}{\partial z_D} \left(v_{tD} = \sum_{j=w,c} v_{jD} \right) = 0 \quad (6)$$

which arises naturally from the incompressibility assumption. Recall that the lateral and transversal fluxes are customarily expressed using Darcy's law, which is given in dimensionless form by the following equations:

$$u_{jD} = -\lambda_{rjD} \left(\frac{\partial p_{jD}}{\partial x_D} + N_{gvj}^{\parallel} \right), \quad v_{jD} = -\kappa \epsilon_{\perp}^{-1} \lambda_{rjD} \left(\frac{\partial p_{jD}}{\partial z_D} - N_{gvj}^{\perp} \right) \quad (7a,b)$$

in which, x_D is aligned with dip direction and increasing upslope and, z_D is dip-normal and increasing downward. Also, the dimensionless mobility is defined as $\lambda_{rjD} = k_{rj}^* k_{rjN} (\mu_1 / \mu_j)$ where k_{rj}^* is phase j 's relative permeability endpoint, $k_{rjN} = f(s_j)$ is phase j 's normalized relative permeability. The dimensionless fluxes define two anisotropic gravity-viscous numbers: (1) a longitudinal $N_{gvj}^{\parallel} = k_h \rho_j g \sin \theta / q_{\mu} \mu_1$ and, (2) a transverse $N_{gvj}^{\perp} = k_h \rho_j g \cos \theta H / q_{\mu} \mu_1 L$ as well as the permeability anisotropy ratio $\kappa = k_v / k_h$ where k_v is the dip-normal absolute permeability. When the dimensionless fluxes are cast into Eq. 5, we arrive at the following equation:

$$\frac{\partial s_{jD}}{\partial t_D} + \frac{\partial}{\partial x_D} \left[-\lambda_{rjD} \left(\frac{\partial p_{jD}}{\partial x_D} + N_{gvj}^{\parallel} \right) \right] + \kappa \epsilon_{\perp}^{-2} \frac{\partial}{\partial z_D} \left[-\lambda_{rjD} \left(\frac{\partial p_{jD}}{\partial z_D} - N_{gvj}^{\perp} \right) \right] = 0. \quad (8)$$

Eq. 8 features the effective shape factor $R_K^2 = \kappa \epsilon_{\perp}^{-2} = (L/H)^2 (k_v/k_h)$. Furthermore, the capillary pressure $p_{cp} = p_2 - p_1$ can be expressed in a dimensionless form as $p_{cD} N_{cv} = (p_{2D} - p_{1D})$ where $p_{cD} = p_c / p_c^*$ and the capillary-viscous number $N_{cv} = p_c^* k_h / q_{\mu} \mu L_x$ in which the characteristic capillary pressure p_c^* is defined from $p_{cp}^* = \int_{s_1} p_c ds_1$ (Zhou, Fayers, and Orr 1994) where $p_c = f(s_1)$ is the fine-scale capillary pressure and $s_1 + s_2 = 1$. At this point, the necessary *primary, phase dimensionless numbers* are defined.

Dimensionality Reduction

Reducing the dimensionality of the problem or the elimination of the dip-normal direction component of Eq. 5 can be achieved in multiple ways (Bear 1972; Bear and Buchlin 1991; Coats, Dempsey, and Henderson 1971; Eldøy, Nygård, and Skjæveland 1984; Guo et al. 2016; Niemi, Bear, and Bensabat 2017; Sorbie et al. 1994; Zapata and Lake 1981). The most obvious, and direct, one is the no vertical flux assumption $v_{jD} = 0$ which implies, from Eq. 7b, for $\lambda_{rjD} > 0$, either $\kappa \epsilon_{\perp}^{-1} \rightarrow 0$ or the following condition:

$$\frac{\partial p_{jD}}{\partial z_D} = N_{gvj}^{\perp} \quad (9)$$

This expression states that the pressure distribution is hydrostatic in the vertical direction - classically, referred to as the Dupuit's assumption. However, notice that it is a result of a prior assumption and not an invoked one. Due to the last point, Phillips (1991) argue that the Dupuit's assumption is not a necessary one for single-phase flows. Alternatively, magnitude analysis reveals that, for term balance to be achieved in Eq. 5, the order of magnitude of the vertical flux must be $O(v_{jD}) \sim \epsilon$ which allows writing the following expression upon direct substitution into Eq. 7b (Lake 1989; Shook, Li, and Lake 1993):

$$\frac{\partial p_{jD}}{\partial z_D} \sim N_{gvj}^\perp - \frac{1}{\lambda_{rjD} R_L^2}. \quad (10)$$

For $R_L \gg 1$ and $\lambda_{rjD} > 0$, the hydrostatic pressure distribution given by Eq. 9 is approached. This implies that the pressure in the vertical direction is stabilized at a much shorter time as compared to that in the lateral direction. It can be seen that this parallel flow argument, or the lubrication approximation as termed by Guo et al. (2016), has led to the Dupuit's assumption. Furthermore, if the flow is completely governed by viscous advection, $N_{gvj}^\perp \rightarrow 0$, and therefore, $\partial p_{jD} / \partial z_D \rightarrow 0$. The latter was invoked as the vertical equilibrium assumption in some studies, see Sorbie et al. (1994), for example.

Eq. 5 can be directly integrated in the dip normal direction to give the following transversely-averaged form:

$$\frac{\partial S_{jD}}{\partial t_D} + \frac{\partial U_{jD}}{\partial x_D} + \epsilon_\perp [v_{jD}]_0^1 = 0 \quad (11)$$

where the transversely-averaged saturation S_{jD} is given by $\int_0^1 s_{jD} dz_D$ and the transversely-averaged lateral flux U_{jD} is $\int_0^1 u_{jD} dz_D$. The dip-normal component is reduced to a sink-source term that vanishes to zero under the conditions of no flux at the bounding surfaces or negligibly small steady-state flux (Bear and Buchlin 1991). Alternatively, using the same argument of parallel flow assumption, the term can be neglected, resulting in:

$$\frac{\partial S_{jD}}{\partial t_D} + \frac{\partial U_{jD}}{\partial x_D} = 0, \quad \phi \frac{\partial S_j}{\partial t} + \frac{\partial U_j}{\partial x} = 0. \quad (12a,b)$$

This discussion shows that transversely-averaged, or the most common name vertically-integrated, models are the general case. However, for those models to be practical, the parallel flow assumption must hold. The parallel flow assumption entails then the Dupuit's assumption. The latter hierarchy is the key point of this section, see Fig. 2. Notice that we have not made any explicit reference to flow regime or flow configuration in the vertical direction. In the next section, we provide a concise qualitative characterization of the analytically tractable regimes and derive the associated governing differential equations.

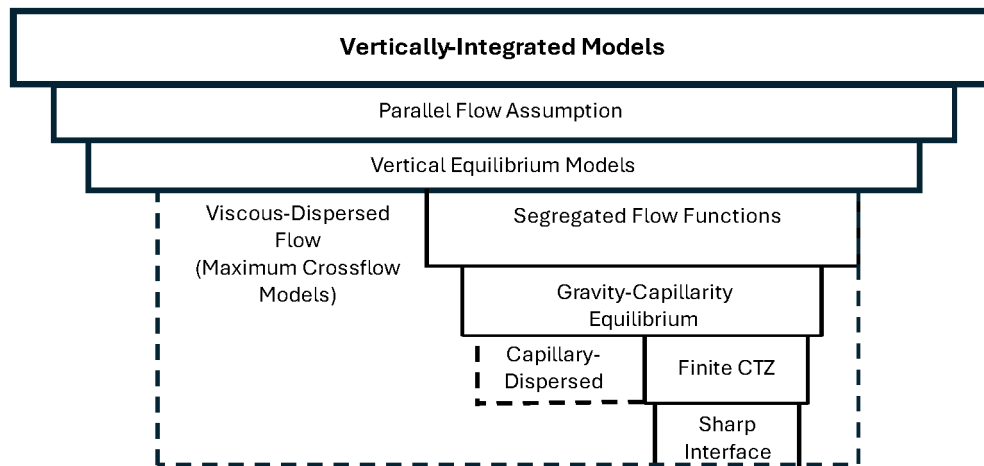


Figure 2—Schematic hierarchy of the different, specific vertically-integrated models.

In the following sections, we assume homogeneity in both fluid and rock properties and defer the discussion on the homogeneous-acting media to the end of the paper. For a detailed discussion on the use of

VE models for heterogeneous models, the reader is referred to [Alamara, Blondeau, Thibeau, et al. \(2025\)](#); [Bandilla, Guo, and Celia \(2017\)](#); [Leahy et al. \(2009\)](#).

Diffuse or Dispersed Flow. Under diffuse or dispersed flow conditions, the saturation is uniform in the vertical direction ($S_j = s_j$). Both high viscous and/or capillary forces cause the latter phenomenon. Therefore, this flow regime can be divided into viscous- and capillary-dispersed regimes. In viscous-dispersed flow, gravitational forces are not strong enough to cause phase segregation, while capillary forces are negligibly small. Viscous-dispersed flow can deservedly be termed Buckley-Leverett flow ([Buckley and Leverett 1942](#)) as it is the canonical example for this theory. For capillary-dispersed flow, the capillary force is much larger than the gravity force, resulting in a uniform saturation in the vertical direction ([Dake 1983](#)). We next derive the general simplified equation for dispersed flows where phase 2 is the lighter, non-wetting phase invading a medium initially saturated by phase 1. Hence, $s_{2m} = 0$, and $S_{2M} = 1 - s_{1irr}$.

The transversely-averaged fluxes can be expressed upon substituting Darcy's law as follows, in dimensional variables for clarity:

$$\begin{aligned} U_1 H &= - \int_0^H \frac{k_{r1}^* k_{r1N}}{\mu_1} \left(\frac{\partial p_1}{\partial x} + \rho_1 g \sin \theta \right) dz = - \frac{k_{r1}^* k_{r1N}}{\mu_1} H \left(\frac{\partial p_1}{\partial x} + \rho_1 g \sin \theta \right). \\ U_2 H &= - \int_0^H \frac{k_{r2}^* k_{r2N}}{\mu_2} \left(\frac{\partial p_2}{\partial x} + \rho_2 g \sin \theta \right) dz = - \frac{k_{r2}^* k_{r2N}}{\mu_2} H \left(\frac{\partial p_2}{\partial x} + \rho_2 g \sin \theta \right). \end{aligned} \quad (13a,b)$$

Even if the pressure distribution can still be represented by hydrostatic distribution, the implicit assumption is that it is negligible, thus a single-valued pressure presents in the vertical direction, i.e., $\partial p_j / \partial z \rightarrow 0$ in the dip-normal direction. The pressure difference between the two phases then equals to the fine-scale, or rock, capillary pressure $p_c = p_2 - p_1$.

From the sum of continuity equation of the phases, we have $U_1 + U_2 = q_t$. Upon substituting the fluxes in the last expression, and solving for $\partial p_2 / \partial x$:

$$\frac{\partial p_2}{\partial x} = - \frac{q_t + (k_{r2}^* k_{r2N} / \mu_2) \rho_2 g \sin \theta - k_{r1}^* k_{r1N} (\mu_1 / \mu_2) \partial p_c / \partial x + k_{r1}^* k_{r1N} (\mu_1 / \mu_2) \rho_1 g \sin \theta}{k_{r1N} (k_{r1}^* / \mu_1) + k_{r2N} (k_{r2}^* / \mu_2)}. \quad (14)$$

Eq. 14 is then substituted in Eq. 13b to give the following expression:

$$U_2 = \frac{k_{r2}^* k_{r2N}}{\mu_2} \left(\frac{q_t + k_{r1}^* k_{r1N} (\mu_1 / \mu_2) (\Delta \rho g \sin \theta - \partial p_c / \partial x)}{k_{r1N} (k_{r1}^* / \mu_1) + k_{r2N} (k_{r2}^* / \mu_2)} \right) \quad (15)$$

where $\Delta \rho = \rho_1 - \rho_2$ is the density difference. Eq. 15 can be substituted into the transversely-averaged differential mass balance given by Eq. 12b, yielding the following equation:

$$\begin{aligned} \frac{\partial}{\partial t} \left(\frac{S_2}{1-s_{1irr}} \right) + \phi (1-s_{1irr}) \frac{\partial}{\partial x} \left[\underbrace{\left(\frac{M k_{r2N}}{k_{r1N} + M k_{r2N}} \right) \left(q_t + \frac{k_{r1}^* k_{r1N} \Delta \rho g \sin \theta}{\mu_1} \right)}_{\text{Non-Linear Diffusion}} \right] \\ = \underbrace{\frac{k_{r2}^*}{\phi (1-s_{1irr}) \mu_2} \frac{\partial}{\partial x} \left[\left(\frac{k_{r1}^* k_{r1N}}{k_{r1N} + M k_{r2N}} \right) \left(\frac{\partial p_c}{\partial x} \right) \right]}_{\text{Non-Linear Advection}} \end{aligned} \quad (16)$$

Eq. 16 is a relatively general equation from which special cases can be derived, see for example ([Blunt and King 1991](#); [Ghanbarnezhad and Lake 2011](#); [Ghanbarnezhad Moghanloo et al. 2015](#); [Noh et al. 2004](#); [Walsh and Lake 1989](#)). The classical Buckley-Leverett theory for viscous-dominated flow with longitudinal (or dip-driven) buoyancy can be obtained by setting $p_c = 0$. The capillary imbibition (non-linear diffusion)

equation in a horizontal model can be obtained by setting $q_i = \theta = 0$, consult [Schmid et al. \(2016\)](#) and the references therein. Eq. 16 can be expressed in dimensionless form as follows:

$$\frac{\partial S_{2D}}{\partial t_D} + \frac{\partial}{\partial x_D} \left[\left(\frac{k_{r2N}}{k_{r1N} + M k_{r2N}} \right) \left((N_b)_{VD} + k_{r1N} \tan \theta \right) \right] = \frac{\partial}{\partial x_D} \left[\left(\frac{k_{r1N} k_{r2N}}{k_{r1N} + M k_{r2N}} \right) \left((Bo)_{VD} \frac{\partial p_{cD}}{\partial x_D} \right) \right] \quad (17)$$

where $S_{2D} = S_2 / (1 - s_{lirr})$, $x_D = x / L_x$, $t_D = k_h k_{r2}^* \Delta \rho g \cos \theta t / \phi (1 - s_{lirr}) \mu_2 L$, and the dimensionless numbers are: (1) dip-driven buoyancy number $(N_b)_{VD} = k_h k_{r1}^* \Delta \rho g \cos \theta / q_i \mu_1$, (2) longitudinal Bond number $(Bo)_{VD} = \Delta \rho g \cos \theta L / p_c^*$, and (3) endpoint mobility ratio $M = (k_{r2}^* / k_{r1}^*) (\mu_1 / \mu_2)$.

Segregated Flow. In segregated flows, buoyancy leads to stratified flow with a distinct interface between the phases. The interface can either be sharp if the gravity segregation dominates over both viscous and capillary forces or diffuse (or smooth) one if the capillary forces are strong enough to establish a capillary transition zone (CTZ) ([Alamara, Blondeau, and Bogdanov 2025](#); [Golding, Huppert, and Neufeld 2013](#); [Nilsen et al. 2016](#); [Nordbotten and Dahle 2011](#)). The sharp-interface is a special case of the more general diffuse (or smooth) interface case on the limit of negligible CTZ. Assuming negligible variations in fluid density in the dip-normal direction, the pressure distribution can be expressed:

$$\begin{aligned} p_2(x, z, t) &= p_T(x, t) + \rho_2 g z \cos \theta, & 0 \leq z \leq h \\ p_1(x, z, t) &= p_T(x, t) + \rho_1 g z \cos \theta - \Delta \rho g h \cos \theta, & h \leq z \leq H \end{aligned} \quad (18)$$

using the same convention of denoting the lighter, non-wetting phase by subscript 2. The thickness h [L] represents that of the lighter phase and, the pressure p_T appearing in Eq. 18 is the pressure at the top z -coordinate of the domain. The transversely-averaged fluxes under the condition of sharp-interface can be expressed as:

$$\begin{aligned} U_2 h &= - \int_0^h \frac{k_h k_{r2}}{\mu_2} \left(\frac{\partial p_2}{\partial x} + \rho_2 g \sin \theta \right) dz \\ &= - \frac{k_{r2}^* k_h}{\mu_2} h \left(\frac{\partial p_T}{\partial x} + \rho_2 g \sin \theta \right), \\ U_1 (H - h) &= - \int_h^H \frac{k_h k_{r1}}{\mu_1} \left(\frac{\partial p_1}{\partial x} + \rho_1 g \sin \theta \right) dz \\ &= - \frac{k_h k_{r1}^*}{\mu_1} (H - h) \left(\frac{\partial p_T}{\partial x} + \rho_1 g \sin \theta - \Delta \rho g \cos \theta \frac{\partial h}{\partial x} \right). \end{aligned} \quad (19a,b)$$

Notice that only the endpoint relative permeabilities are relevant under perfect gravity segregation. We will study this later.

From the condition of incompressibility, we have:

$$\frac{\partial}{\partial x} ((H - h) U_1 + h U_2) = 0 \rightarrow (H - h) U_1 + h U_2 = q_i H. \quad (20a,b)$$

Notice that q_i can vary with time. Substituting Eqs. 19a and 19b in Eq. 20b and solving for $\partial p_T / \partial x$ to eliminate p_T from the transversely-averaged fluxes, gives the following:

$$\frac{\partial p_T}{\partial x} = - \frac{q_i + \left(k_h (k_{r2}^* / \mu_2) (h / H) \rho_2 g \sin \theta + k_h (k_{r1}^* / \mu_1) (1 - h / H) \rho_1 g \sin \theta - k_h (k_{r1}^* / \mu_1) (1 - h / H) \Delta \rho g \cos \theta \frac{\partial h}{\partial x} \right)}{k_h (k_{r1}^* / \mu_1) (1 - h / H) + k_h (k_{r2}^* / \mu_2) (h / H)} \quad (21)$$

which can be substituted into Eq. 19a resulting in complete elimination of p_T :

$$U_2 = \frac{k_{r2}^*}{\mu_2} \left(\frac{h}{H} \right) \left(\frac{q_t + k_h (k_{r1}^* / \mu_1) (1 - h/H) \Delta \rho g (\sin \theta - \cos \theta \frac{\partial h}{\partial x})}{(k_{r1}^* / \mu_1) (1 - h/H) + (k_{r2}^* / \mu_2) (h/H)} \right) \quad (22)$$

This equation can then be substituted into Eq. 12b with $S_2 = \int_0^h s_j dz / H = h(1 - s_{lirr}) / H$ as a result of the sharp-interface assumption to arrive at the following final equation:

$$\begin{aligned} \frac{\partial}{\partial t} \left(\frac{h}{H} \right) + \frac{1}{\phi(1-s_{lirr})} \frac{\partial}{\partial x} \left[\underbrace{\left(\frac{M(h/H)}{(1-h/H)+M(h/H)} \right) \left(q_t + \frac{k_h k_{r1}^* \Delta \rho g \sin \theta}{\mu_1} (1 - h/H) \right)}_{\text{Non-Linear Advection}} \right] \\ = \underbrace{\frac{k_h k_{r2}^*}{\phi(1-s_{lirr})\mu_2} \frac{\partial}{\partial x} \left[\left(\frac{(1-h/H)(h/H)}{(1-h/H)+M(h/H)} \right) (\Delta \rho g \cos \theta \frac{\partial h}{\partial x}) \right]}_{\text{Non-Linear Diffusion}} \end{aligned} \quad (23)$$

Eq. 23 is a relatively general form for gravity-segregated flows in homogeneous media, as it accounts for multiple driving forces. Various specific analytical and semi-analytical solutions have been proposed based on it (Alamara, Blondeau, and Bogdanov 2023a; Ekran 1992; Fayers and Muggeridge 1990; Golding et al. 2011; Hesse, Orr, and Tchelep 2008; Huppert and Woods 1995; De Loubens and Ramakrishnan 2011; Lyle et al. 2005; Martin 1958; Mitchell and Woods 2006; Nijjer et al. 2022; Pegler, Huppert, and Neufeld 2014; Pritchard, Woods, and Hogg 2001; Vella and Hupper 2006; Zhao et al. 2014), and recently utilized for studying the periodic hydrogen injection-withdrawal in aquifers by Whelan and Woods (2025). It can be written in a dimensionless form as follows:

$$\frac{\partial h_D}{\partial t_D} + \frac{\partial}{\partial x_D} \left[\left(\frac{h_D}{1+(M-1)h_D} \right) \left((N_b)_{GS}^{-1} + (1-h_D) \left(N_\theta - \frac{\partial h_D}{\partial x_D} \right) \right) \right] = 0 \quad (24)$$

where the dimensionless distance is $x_D = x/L_x$, dimensionless interface thickness $h_D = h/H$, and the dimensionless time $t_D = k_{r2}^* k \Delta \rho g \cos \theta H t / \phi(1-s_{wirr})\mu_2 L_x^2$. The dimensionless numbers obtained are: (1) endpoint mobility ratio $M = (k_{r2}^* / k_{r1}^*) (\mu_1 / \mu_2)$, (2) dimensionless buoyancy number $(N_b)_{GS} = k_{r1}^* k_h \Delta \rho g \cos \theta H / q_t \mu_1 L_x = N_{gvw}^\perp - N_{gvc}^\perp$, and (3) dimensionless slope number $N_\theta = (L_x/H) \tan \theta$ reflecting the ratio of $(N_{gvw}^\parallel - N_{gvc}^\parallel)$ to $(N_{gvw}^\parallel - N_{gvc}^\parallel)$. In the absence of injection flux, $(N_b)_{GS}^{-1} \rightarrow 0$ and the equation becomes that studied by Hesse et al. (2008). Finally, we mention that flows which can be described by Eq. 24 can be termed Dietz flow (Dietz 1953) as opposite to the Buckley-Leverett flow.

Generalization. A careful examination of Eq. 16 and 23 reveals that they share the same mathematical structure. Consequently, conventional reservoir simulators—which typically assume dispersed flow—can, in principle, be adapted to simulate segregated flows, provided that the following functions are appropriately defined:

$$K_{r1} = k_{r1}^* (1 - h/H) \quad , \quad K_{r2} = k_{r2}^* (h/H) \quad , \quad P_c = \Delta \rho g h \cos \theta \quad , \quad S_2 = (1 - s_{lirr}) (h/H). \quad (25a,b,c,d)$$

Furthermore, it is possible to generalize the results of Eq. 16 and 23 by utilizing a convex combination of the dispersed and segregated, sharp-interface flow models:

$$\begin{aligned} K_{rjN}^{eff} &= \alpha K_{rjN} + (1 - \alpha) k_{rjN} \\ P_c^{eff} &= \alpha P_c + (1 - \alpha) p_c \end{aligned} \quad (26a,b)$$

Therefore, we have:

$$\begin{aligned}
& \frac{\partial}{\partial t} \left(\frac{S_2}{1-s_{lirr}} \right) + \underbrace{\frac{1}{\phi(1-s_{lirr})} \frac{\partial}{\partial x} \left[\left(\frac{MK_{r2N}^{eff}}{K_{r1N}^{eff} + MK_{r2}^{eff}} \right) \left(q_t + \frac{k_h k_{r1}^* K_{r1N}^{eff} \Delta \rho g \sin \theta}{\mu_1} \right) \right]}_{\text{Non-Linear Advection}} \\
& = \underbrace{\frac{k_h k_{r2}^*}{\phi(1-s_{lirr}) \mu_2} \frac{\partial}{\partial x} \left[\left(\frac{MK_{r1N}^{eff} K_{r2N}^{eff}}{K_{r1N}^{eff} + MK_{r2}^{eff}} \right) \left(\frac{\partial P_c^{eff}}{\partial x} \right) \right]}_{\text{Non-Linear Diffusion}}.
\end{aligned} \tag{27}$$

A value of $\alpha = 0$ represents the case of completely dispersed flow, whilst $\alpha = 1$ represents the case of perfectly segregated flow with sharp-interface. Despite this generalization approach had been implemented in the industry standard commercial simulation ECLIPSE (SLB 2024a), it has been rarely reported and discussed in literature. Given that α changes with the flow condition from 0 to 1, we expect that it is primarily a function of a buoyancy number. Another interesting point to be verified is that α is principally not a phase-dependent parameter.

This section is aimed at providing transparent derivation of the flow equations avoiding the common shortcuts and attempting to be as general as possible. Key point to note is that the pseudo-capillary pressure P_c given by Eq. 25c has nothing to do with the fine-scale capillary pressure but arose due to the mathematically diffusive nature of the thickness-driven flow due to gravity. Following the derivation and the inspectional comparison clarifies this point. The authors wish to emphasize that the assumption of incompressibility is non-restrictive for reservoir simulator applications but made here for the purpose of clarity. However, the assumption of negligible fluid properties variations in the dip-normal direction is an obvious limitation (Andersen, Gasda, and Nilsen 2015; Andersen and Nilsen 2018). Linearized formulations show the feasibility of assuming vertically negligible variations in fluid properties (Alamara, Blondeau, Thibeau, et al. 2025). So far, the derivations are provided for the drainage case; the reader is referred to a number of studies that consider residual trapping and hysteresis modeling (Alamara, Blondeau, and Bogdanov 2023b; Alamara, Bogdanov, and Blondeau 2025; Doster, Nordbotten, and Celia 2012, 2013; Goda and Sato 2011; Hesse et al. 2008; Juanes et al. 2006; Kochina, Mikhailov, and Filinov 1983; Nilsen et al. 2016).

Mapping Flow Regimes

The previous section concentrates on deriving and scaling the reduced-dimensionality flow equations across the whole domain. In this section, we turn our focus to the flow regime and its impact on the phase distribution in the vertical direction which depends on the relative importance of the forces to each other. We hypothesize that the following dimensionless numbers can determine the local flow regime, and thus, the distribution of the phases in the vertical direction:

$$N_b = \frac{k_v \Delta \rho g \cos \theta}{q_t \mu_1} = \frac{\kappa}{\epsilon_{\perp}} (N_{gv1}^{\perp} - N_{gv2}^{\perp}), \quad N_c = \frac{p_c^* k_v}{q_t \mu_1 H} = \frac{\kappa}{\epsilon_{\perp}} N_{cv}, \quad Bo = \frac{\Delta \rho g \cos \theta}{p_c^* / H} = \frac{N_b}{N_c}. \tag{28a,b,c}$$

We refer to the first number as the "intrinsic" buoyancy number which describes the importance of a characteristic gravity-driven flux as relative to total pressure-driven advection, or displacement flux. The second measures the importance of capillary forces relative to that flux. The ratio of the first two numbers defines the Bond number which estimates the segregative buoyancy force relative to the diffusive capillary force. Those numbers do not feature the aspect ratio e and can be thought of as "interdomain" versions of the large-scale numbers featuring this ratio.

The flow map in Fig. 3 shows the different flow regimes for our preliminary large-scale estimates. We will refine the boundaries later in this work focusing on the sharp-interface case. The figure plots N_c vs N_b with the diagonal lines representing $\log Bo$ with dimensionless simulation models' results juxtaposed on the same figure. The flow map can be divided into two primary regions: (1) dispersed flow for $N_b \ll 1$

and, (2) segregated flow for $N_b \gg 1$. The dispersed flow region can either be viscous- or capillary dispersed depending on the strength of the capillary force compared to the viscous force. In this region, simulation models can be confidently reduced in dimensionality and simulated with the fine-scale or the rock flow functions (relative permeability and capillary pressure). On the other hand, the segregated flow region, can be divided into: (1) a sharp-interface flow region where capillary force is not strong enough to establish an appreciable capillary transition zone, i.e., $Bo \gg 1$, and (2) a region where the capillary force compensates the gravity and establish a gravity-capillarity equilibrium (GCE) but gravity is strong enough to keep a region of free lighter phase at maximum saturation, i.e., $Bo \sim 1$, and (3) a region where flow is still segregated but capillary force is strong enough to disperse the flow vertically, eliminating the zone where the lighter phase exists at its maximum, i.e., $Bo \ll 1$. The reader is also referred to Guo et al. (2016) for a study of the different flow regimes but based on the sharp-interface vertical equilibrium formulation.

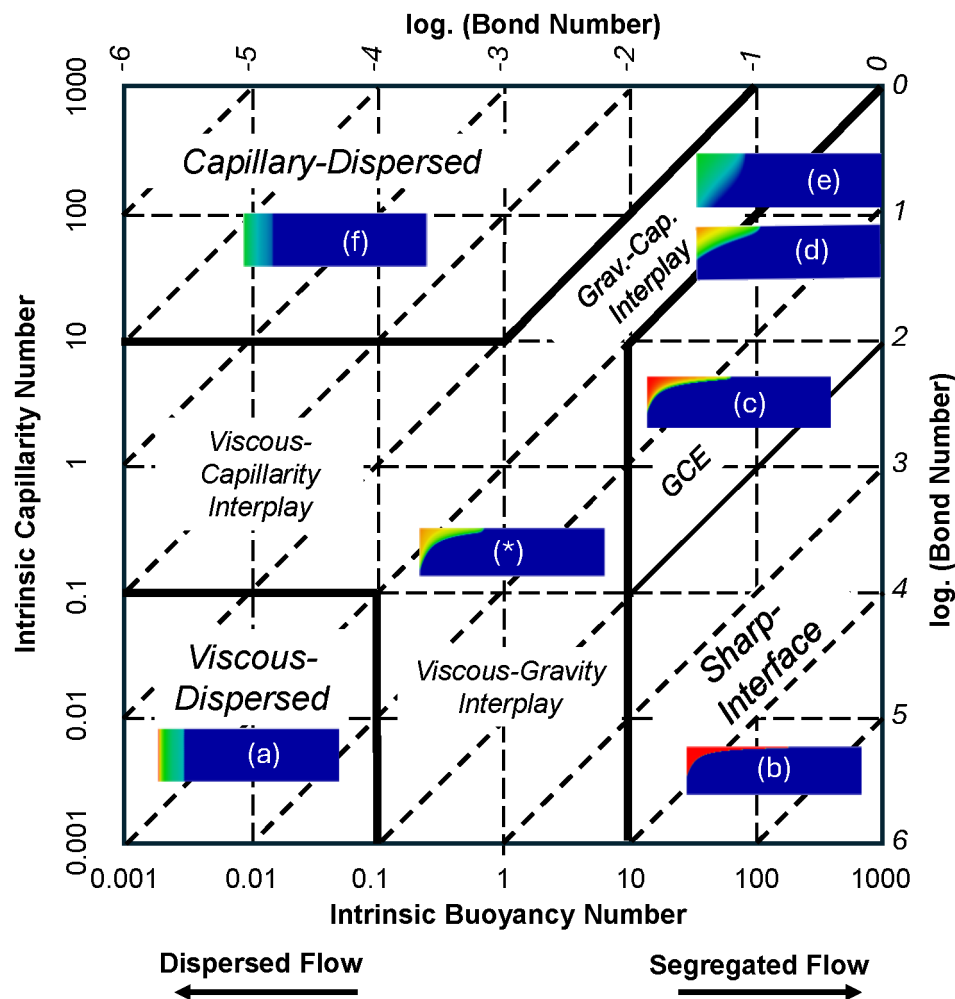


Figure 3—Preliminary flow map illustrating the different flow regimes. (a) viscous-dispersed flow regime $N_b \rightarrow 0$, $N_c \rightarrow 0$, (b) Gravity-segregated, sharp-interface flow regime $N_b = 10$, $N_c \rightarrow 0$, $B_o \rightarrow \infty$, (c) Gravity-segregated with gravity-capillarity equilibrium $N_b = 2$, $N_c = 0.4$, $B_o = 5$, (d) Gravity-segregated with thick capillary transition zone $N_b = 2$, $N_c = 2$, $B_o = 1$, (e) Segregated with important capillary dispersion $N_b = 2$, $N_c = 4$, $B_o = 0.5$, (f) Capillary-dispersed flow $N_b \rightarrow 0$, $N_c = 30$, $B_o \rightarrow 0$.

Role of Capillarity-Gravity Equilibrium

Capillary pressure has the impact of finite dispersion both longitudinally (along the displacement's main path) and transversely (in the normal direction to the displacement's main path). The longitudinal impact is introduced to the vertically-integrated model through simple addition of the fine-scale capillary pressure

to the sharp-interface pseudo-capillary pressure $P_c^{GCE} = P_c^{SI} + p_c \sim P_c^{SI}$ (Martin 1968). On the other hand, the transverse impact is introduced through assuming instantaneous gravity-capillarity equilibrium (Coats et al. 1971; Golding et al. 2013; Nordbotten and Dahle 2011) which can be considered as the maximum transverse capillary dispersion. The latter is embedded through the vertically-averaged relative permeability and achieved by assuming different levels of contact, imposing the capillary rise on top of each level, and calculating the corresponding relative permeability at each point then, calculating the integrals given by:

$$S_2 = \frac{\int_0^h s_2 dz + \int_h^{h+\Delta h_{ctz}} s_2 dz}{H} = \frac{(1-s_{lirr})h + \int_h^{h+\Delta h_{ctz}} s_2 dz}{H},$$

$$K_{r1} = \frac{k_{r1}^*(H - (h + \Delta h_{ctz})) + \int_h^{h+\Delta h_{ctz}} k_{r1} dz}{H}, \quad (29a,b,c)$$

$$K_{r2} = \frac{\int_0^h k_{r2} dz + \int_h^{h+\Delta h_{ctz}} k_{r2} dz}{H} = \frac{k_{r2}^* h + \int_h^{h+\Delta h_{ctz}} k_{r2} dz}{H}.$$

For the fine-scale relative permeability and capillary pressure shown in Fig. 4 (c: CO₂, w: water), a single point calculations example is shown in Fig. 5, the successive application gives the curves shown in Fig. 6. The sharp-interface is a special case with $\Delta h_{ctz} \rightarrow 0$. Notice the subtle curvature in the upscaled relative permeability curves in Fig. 6 which introduces a self-sharpening effect to the interface as reported by Nordbotten and Dahle (2011) using 1D models and Alamara, Blondeau, and Bogdanov (2025) using 2D models. This self-sharpening effect can be readily understood by observing the reduced phase relative permeability as compared to the sharp-interface straight-lines. While its impact might be neglected during injection, the presence of CTZ cannot be treated so in the long-term migration given the expected small thickness of the lighter phase plume. The reader is referred to the last two studies cited for further investigation.

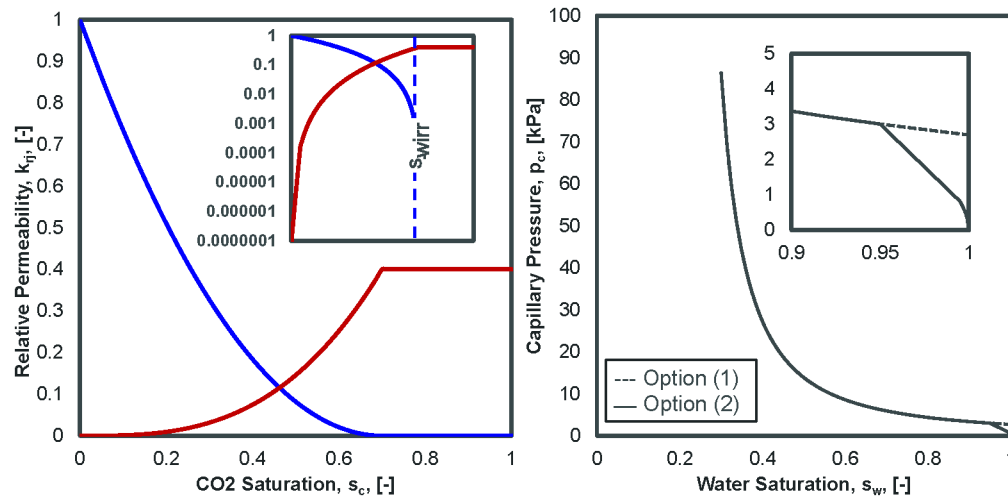


Figure 4—Fine-scale flow functions (relative permeability and capillary pressure). We used two types of capillary entry pressures. No impact has been observed on the averaged-flow functions.

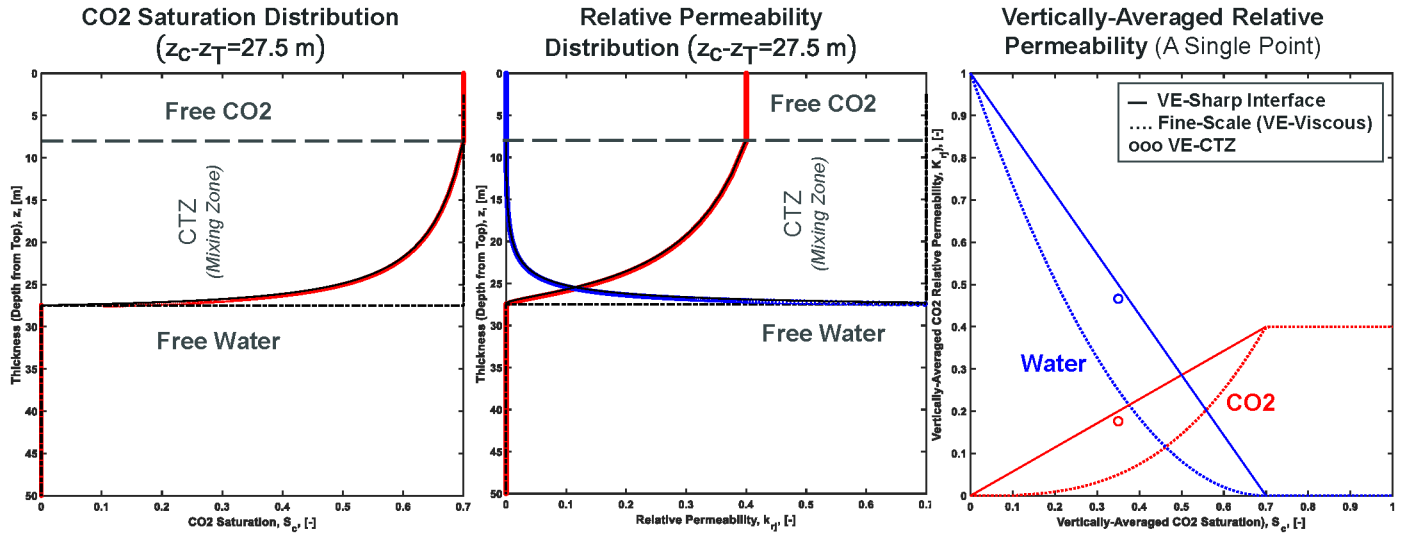


Figure 5—Example calculations of VE-CTZ vertically-averaged relative permeability curves.

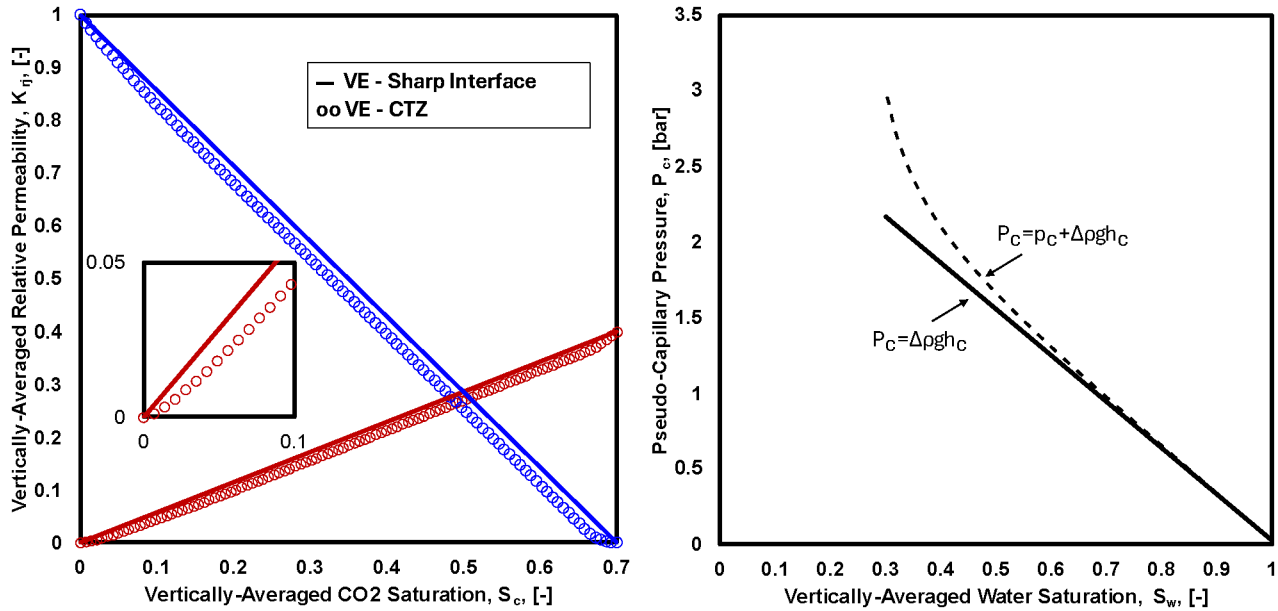


Figure 6—VE-CTZ vertically-averaged relative permeability and pseudo-capillary pressure.

Our purpose here is to determine the time required to achieve gravity-capillarity equilibrium as a measure of the suitability of assuming its instantaneous occurrence. [Fries & Dreyer \(2008\)](#) derived an expression for the time required for a wetting phase imbibing into a capillary tube to reach 99% of its maximum rise restrained by gravity, denoted by t_s . They also provided a porous media analogue:

$$t_s \cong 3.62 \frac{a}{b^2}; a = \frac{\sigma R \cos \theta}{4\mu} \equiv \frac{2\sigma \cos \theta_c}{\phi \mu} \frac{k}{R}; b = \frac{\rho g R^2 \sin \theta}{8\mu} \equiv \frac{\rho k g \sin \theta}{\phi \mu}. \quad (30)$$

where σ is the surface tension [Mr^{-2}], R is the pore radius [L], μ is the wetting-phase viscosity [$ML^{-1}t^{-1}$], θ_c is the contact angle $[-]$, ϕ is the porosity $[-]$, k is the permeability [L^2], ρ is the wetting-phase density [ML^{-3}], g is the gravitational acceleration [Lt^{-2}], and θ is the dip angle $[-]$. We will make the following empirical modifications to produce a representative porous-media version:

1. We substitute for $2\sigma \cos \theta_c / R$ by $(p_c(s_{1irr}) - p_d)$ where p_d is the capillary displacement pressure. Basically, this means making the time estimate with respect to the most stringent condition (longest time).
2. Instead of only the denser phase density coming from assuming $\rho_1 \gg \rho_2$, we use $\Delta\rho = \rho_1 - \rho_2$.
3. Instead of relying only on the water viscosity, we make a physical inference backed by the following logical: notice that $\mu = \mu_1$ is on the numerator. As it reflects on the mobility, in a porous media, we expect μ_1/k_{r1}^* . Then, the other missing factor is the lighter phase mobility μ_2/k_{r2} . The theory of the fluid flow in porous media shows that the overall mobility should appear similar to $\left(\frac{k_{r1}^*}{\mu_1}\right)\left(1 + \frac{k_{r1}^*}{k_{r2}^*} \frac{\mu_2}{\mu_1}\right)$.

For which, the representative μ must be $(\mu_1 k_{r2}^* + k_{r1}^* \mu_2) / k_{r1}^* k_{r2}^*$.

4. In account for the tortuosity that is not accounted for in the permeability expression, we will multiple the stabilization time t_s by the tortuosity factor τ . The time to gravity-capillarity equilibrium t_{GCE} is thus $t_{GCE} = t_s \tau$.

Points (1) to (4) above yield the following expression for the time to 99% gravity-capillarity equilibrium t_{GCE} in the vertical direction can be expressed by the following - take $\sin \theta = 1$:

$$a = \frac{k_{r1}^* k_{r2}^* (p_c(s_{1irr}) - p_d) k_v}{\phi (\mu_1 k_{r2}^* + \mu_2 k_{r1}^*)}, \quad b = \frac{k_{r1}^* k_{r2}^* k_v \Delta\rho g \sin \theta}{\phi (\mu_1 k_{r2}^* + \mu_2 k_{r1}^*)}. \quad (31)$$

Therefore:

$$t_{GCE} \cong 3.62 \tau \frac{(p_c(s_{1irr}) - p_d) / \Delta\rho g}{\Delta\rho g k_v} \phi \left(\frac{\mu_1 k_{r2}^* + \mu_2 k_{r1}^*}{k_{r1}^* k_{r2}^*} \right). \quad (32)$$

But the maximum capillary transition zone thickness is $h_{CTZ} = (p_c(s_{2irr}) - p_d) / \Delta\rho g$, which gives finally:

$$t_{GCE} \cong 3.62 \tau \left(\frac{\mu_1 k_{r2}^* + \mu_2 k_{r1}^*}{k_{r1}^* k_{r2}^*} \right) \left(\frac{\phi h_{CTZ}}{\Delta\rho g k_v} \right). \quad (33)$$

The prefactor 3.62 can be treated as a fitting parameter. Fig. 7 depicts a time evolution towards gravity capillarity equilibrium of model with initial sharp-interface separating the fluids. The last subplot features a comparison with the analytical $t \rightarrow \infty$ solution i.e., the classical gravity-capillarity equilibrium (GCE) calculations. Also, notice that the semi-analytical estimation proved accurate enough or at least useful as a scale estimate. In fact, the rate of rise is extremely slow above 90% of the maximum rise. We tested the accuracy of the semi-analytical estimate presented in Eq. 33 for predicting time to GCE within a simulation framework. By systematically varying vertical permeability over a wide range (1 – 10,000 md), we assessed the analytical estimate against numerical simulation results. We have also changed the density difference and the porosity at two checkpoints. The results in Fig. 8 illustrate the alignment between the analytical estimate and numerical results across a diverse range of cases.

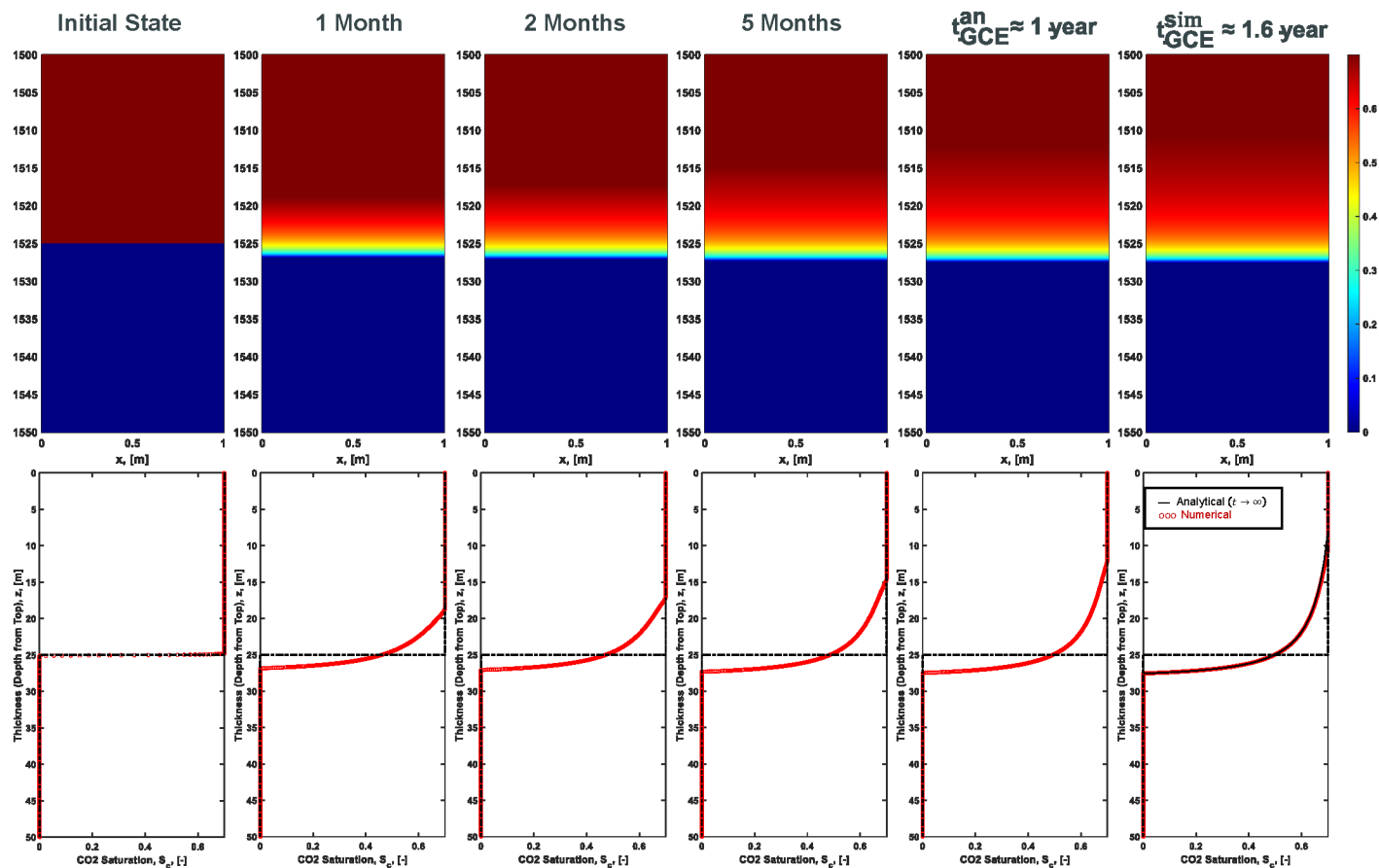


Figure 7—Phase redistribution until gravity-capillary equilibrium. Solid black line represents analytical ($t \rightarrow \infty$) solution.

Particularly, for VE with diffuse (or smooth) interface simulation with CTZ, it can be seen from the numerical values in Fig. 8 that assuming instantaneous gravity-capillarity equilibrium makes a reasonable assumption considering the simulation time scale for CO₂ storage. If the vertical permeability is roughly greater than 100 md, under typical conditions postulated earlier, GCE occurs during the injection time scale, making the long-term utilization even more suitable.

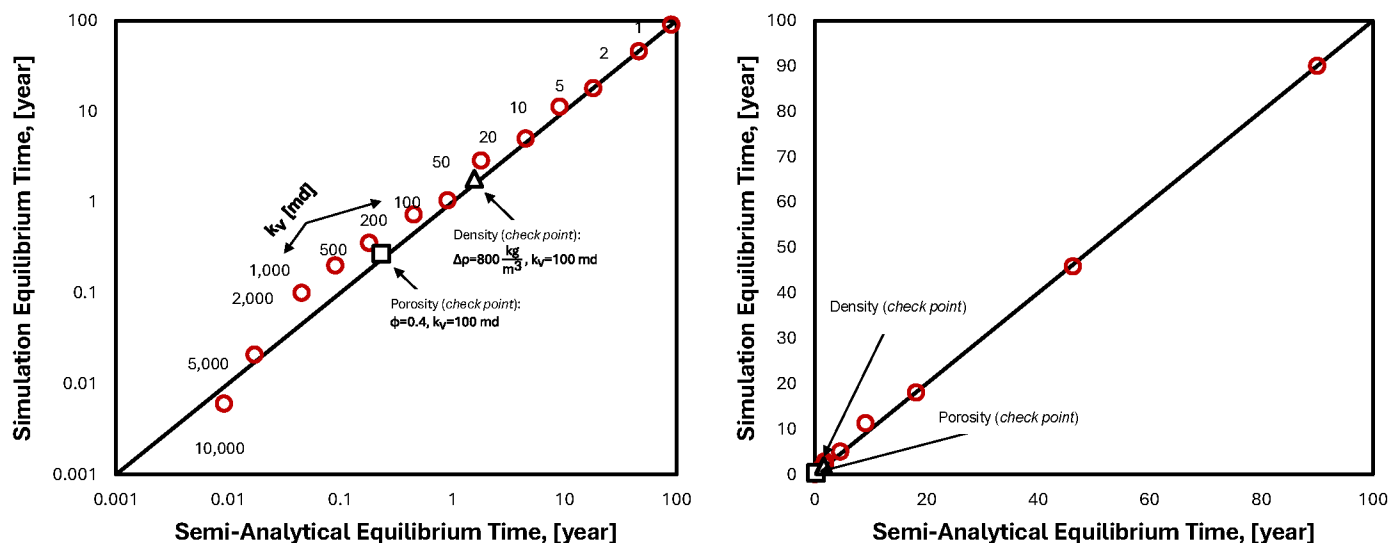


Figure 8—Comparison between semi-analytical and detailed simulation equilibrium time (log and normal).

Favorable Flow Conditions for Segregated Flow VE

From the previous discussion, it becomes clear that no matter whether the flow is with an important capillary transition zone or with a negligible one, it must be characterized as segregated first to apply the diffuse or sharp-interface approaches. Otherwise, the flow is either dispersed or in a transition zone between the different flow regimes, and alternative approaches must be used. For the purpose of determining the conditions under which the flow is segregated, a number of vertically-refined numerical models spanning a relatively wide range of different vertical permeability values, total injection rate, and density difference were generated. The base case model's properties tabulated in Tab. 1, neglecting the capillary pressure. The intrinsic (or interdomain) buoyancy number $N_b = k_v \Delta \rho g \cos \theta / q_t \mu_1$, previously introduced, is used as a correlation variable and to distinguish between the different cases. If the flow is segregated, both the numerically averaged relative permeabilities as well as the numerically calculated pseudo-capillary pressure must coincide with the straight-line relationships expected from the analytical sharp-interface model derived earlier. The numerically calculated pseudo-capillary pressure is simply the difference between the extrapolated phases pressure at the top of the simulation domain. We focus on showing results of the more stringent case of injection as the flow is solely driven by buoyancy in its absence (long-term migration) - which makes the vertical equilibrium with segregated flow functions a natural modeling approach.

Table 1—Base case model's properties for dimensionless analysis.

Dimensions × Grid(x-z)	10 km * 100 m * 50 m × [1000, 1,50]
Dip angle	0 deg.
Permeability/porosity [isotropic and homogeneous]	100 md/ 20%
Base case dimensionless interdomain gravity-viscous number ($N_b = k_v \Delta \rho g / q_t \mu_w$)	2
Injection rate	0.01 PV/year
Injection period/simulation time	10 years/1000 years
Mobility ratio ($M = k_{rc}^* \mu_w / \mu_c$)	1.8
Viscosity ratio (μ_w / μ_c)	13.5
In situ density difference	436 kg/m ³
Irreducible water saturation	0.3

Fig. 9 shows some simulated cases based on varying vertical permeability. Notice that, unsurprisingly, the numerically averaged relative permeability tends towards the straight-line (segregated flow at vertical equilibrium conditions in a homogeneous-acting medium) as the gravity impact increases. The change in the slope in the transitional cases ($N_b = 0.02, 0.2$) reflects the local forces, i.e., points with higher CO₂ saturation are closer to the viscous-dominated (dispersed flow) regions and thus, they are closer and eventually identical to the fine-scale relative permeability. In those numerical experiments, a value of $N_b > 2$ seems reasonable as a cut-off value for the suitability of assuming segregated flow.

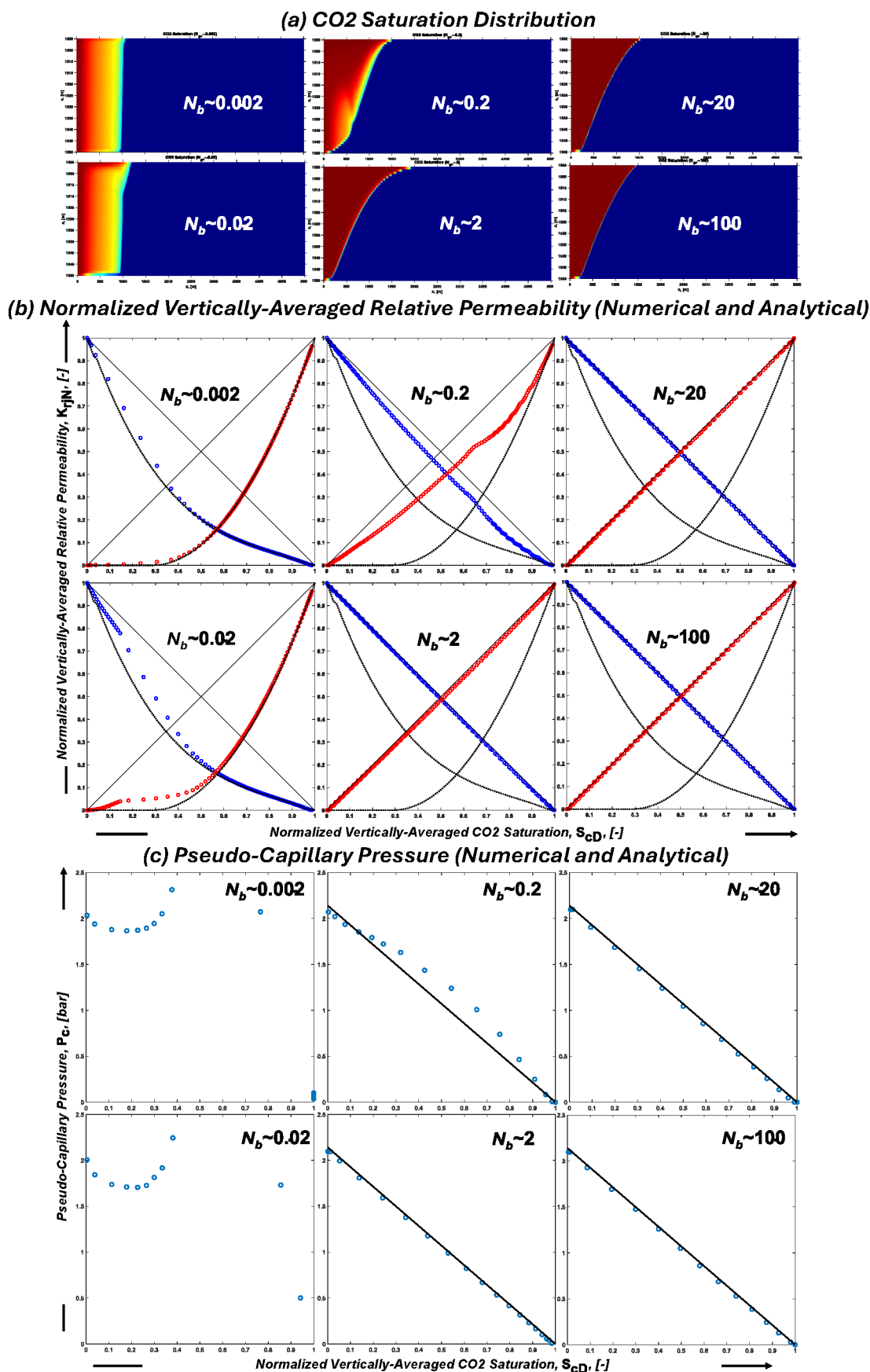


Figure 9—Saturation distributions from different numerical experiments (top), normalized vertically-averaged relative permeability (middle), and pseudo-capillary pressure (bottom). In all subfigures, the dots represent the numerical results.

The other two measures are the profile linearity coefficient m and the absolute relative difference of the plume extent. The first is defined by the following expression:

$$m = 2 \int_0^1 |S_{cD}^{2D} - S_{cD}^{VE-SI}| dS_{cD}^{VE-SI} \quad (34)$$

where S_{cD}^{2D} is the normalized vertically-averaged CO₂ saturation obtained from the vertically-refined 2D model (superscript 2D) and S_{cD}^{VE-SI} is that obtained from the vertical equilibrium model with sharp-interface flow functions (superscript VE-SI). A value of 0 indicates the two are perfectly identical while a value of 1 means that they are completely uncorrelated, see Fig. 10 which plots the results of sampled numerical experiments.

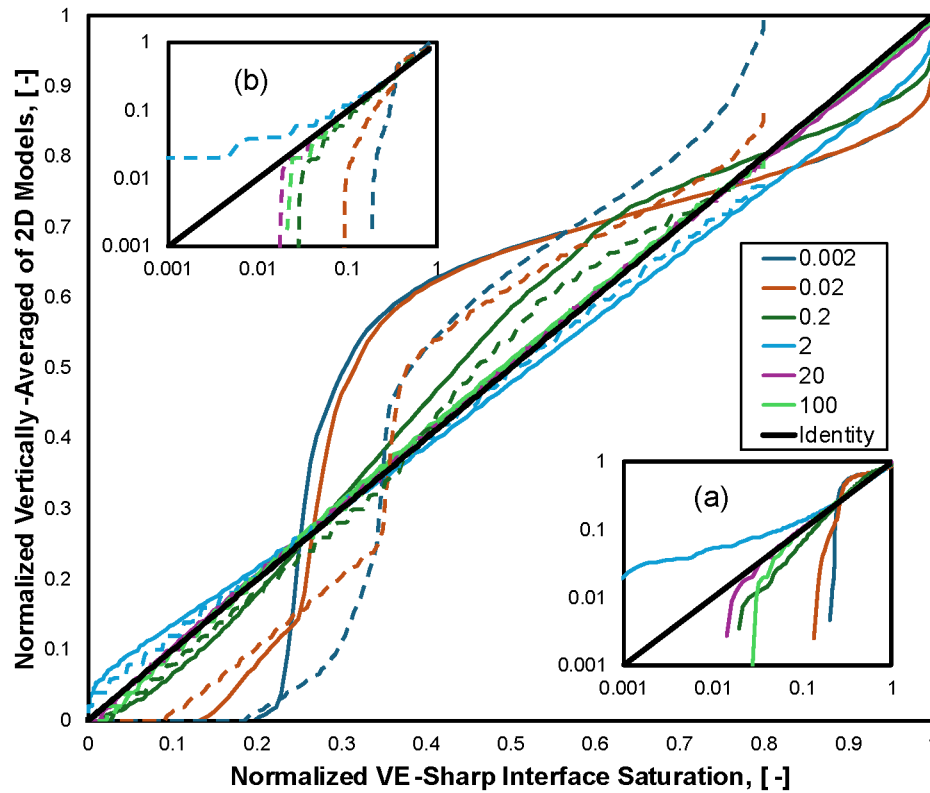


Figure 10—Normalized vertically-averaged saturations (from 2D and VE models) identity plot. The normalized integral of the area between the identity line and the different curves represents the profile linearity coefficients.

The other measure is defined by the following:

$$\Delta = \frac{|x_f^{2D} - x_f^{VE-SI}|}{x_f^{2D}} \quad (35)$$

where x_f is the plume tip distance, superscripts are as in the previous definition. A plot of the two measures against the buoyancy number N_b is shown in Fig. 11. In fact, despite the level of grid refinement we used, both measures are still prone to numerical diffusion. However, the first measure is less prone to it as compared to the second.

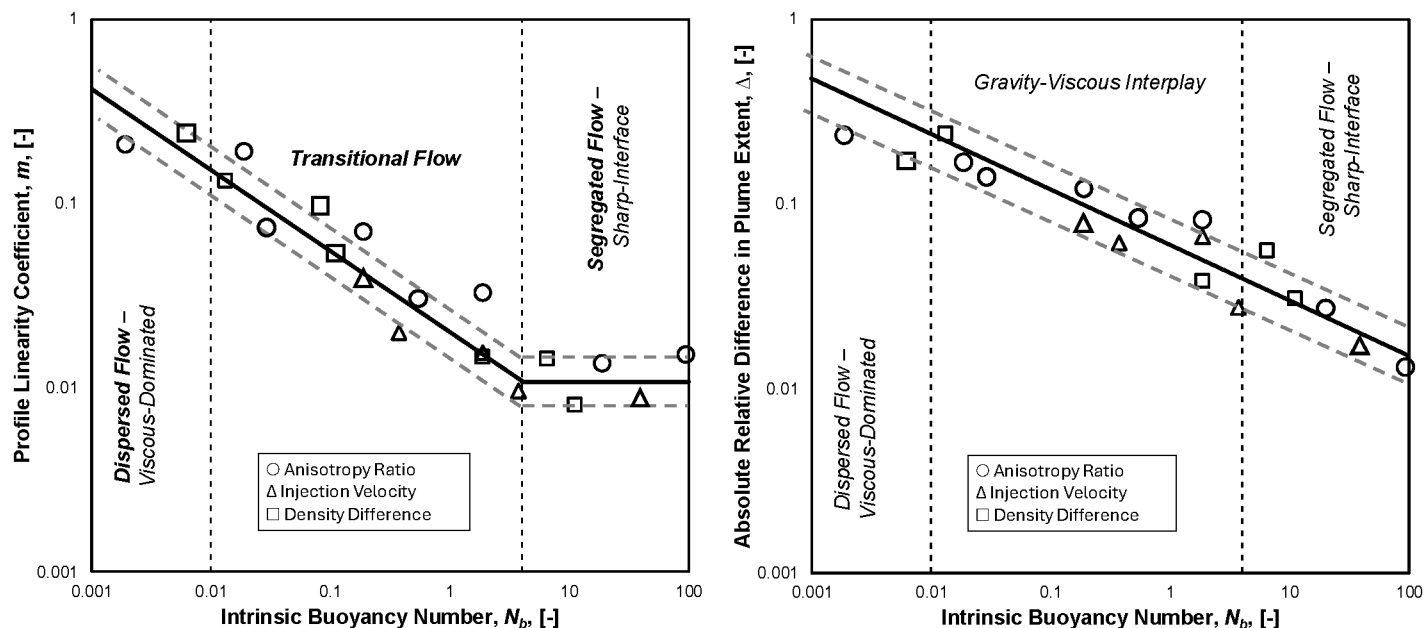


Figure 11—Profile linearity coefficient (left) and the absolute relative difference in the plume extent predictions (right) plotted as function of the intrinsic buoyancy number for the different numerical experiments.

The results of Fig. 11 show that an $N_b \approx 3$ is a reasonable cut-off to consider the flow as segregated. Interestingly, both measures decline as a power-law of N_b . We mention that other measures are available in the literature (Becker et al. 2022; Coats et al. 1971; Fayers and Muggeridge 1990; Nordbotten and Celia 2006; Nordbotten and Dahle 2011).

Estimating α as a Function of N_b

We leveraged the numerical models' results to obtain a correlation between the weighting-factor α , which allows generalizing the models as discussed earlier, as a function of N_b . By varying α until a decent match in the vertically-averaged saturation profile is obtained - see Fig. 12, then correlating it to the model's N_b , we arrived at the following simple empirical correlation:

$$\alpha \approx \left(\frac{4}{3}\right) N_b^{1/4}, \quad N_b \leq \left(\frac{4}{3}\right)^4. \quad (36)$$

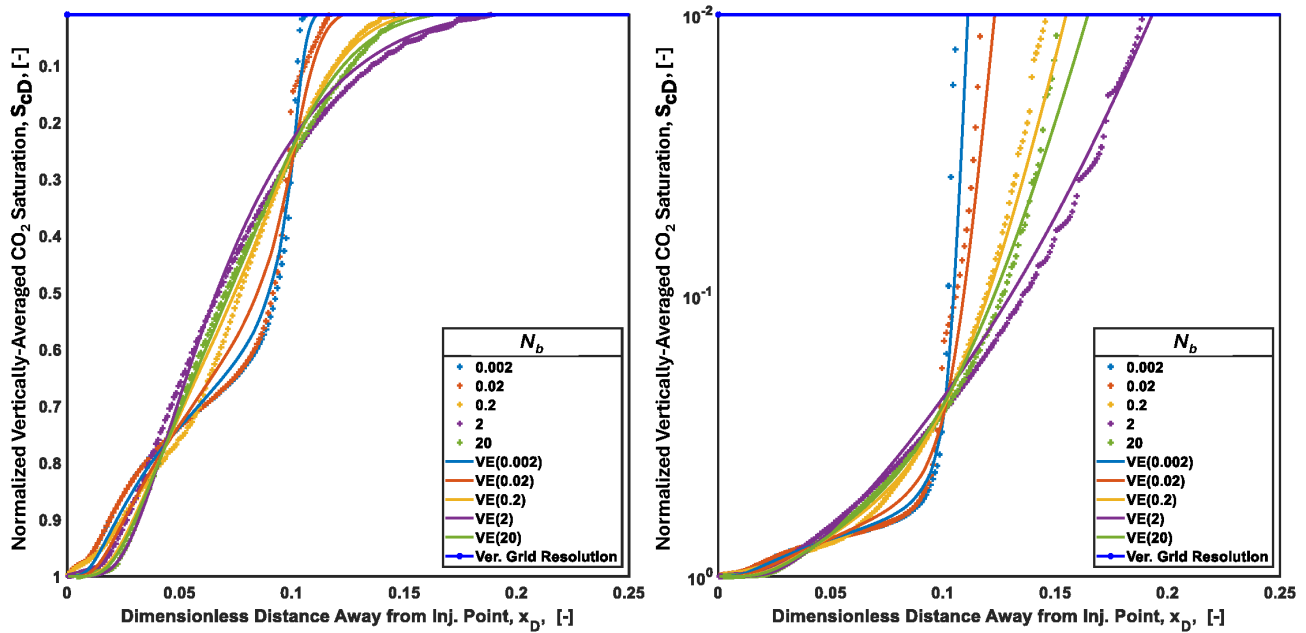


Figure 12—Normalized vertically-averaged saturation in cartesian (left) and semi-log (right) coordinates of the vertically-refined models and the tuned VE models.

Guide Map

In light of what has been presented thus far, a guide map was constructed and is depicted in Fig. 13 alongside a chart of the corresponding α values for each region. The green regions represent those where the vertically-integrated models can be used while the red regions are those involving complex forces interplay that, generally speaking, require fully-resolved models. Different α values can be used depending on the prevailing condition across the models. For instance, a low value is expected near the injection well whilst higher values are expected away from it.

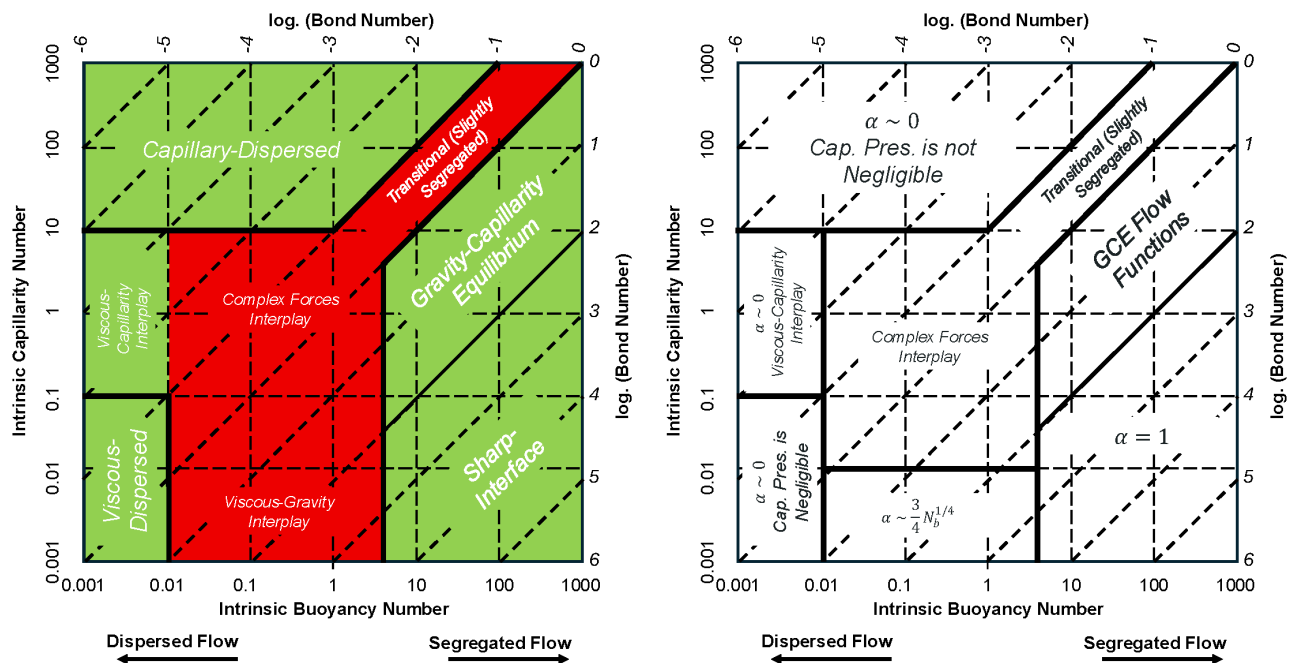


Figure 13—Refined flow regime map where the vertically-integrated approach is reasonable (left) and the corresponding α factor for each region (right).

Sensitivity of Fine-Scale Relative Permeability on Sharp-Interface VE Models

Sharp-interface models give no account for the curvature of the relative permeabilities as described earlier due to the assumption that the mixing zone is of negligible thickness and the phases flow as segregated physical bodies concurrently. To examine if the relative permeability curvature impacts the adequacy of the sharp-interface models, we generated several relative permeabilities as per the following: (1) only the CO₂ fine-scale relative permeability curvature is changed, water relative permeability is straight-line, (2) only the water fine-scale relative permeability curvature is changed, CO₂ relative permeability is straight-line, (3) both CO₂ and water relative permeabilities curvature change. To mention that VE models with smooth interface (finite capillary transition) are obviously impacted by the shape of the fine-scale curves.

The models' properties are the same as the base case of the previous sections with $N_b \sim 2$. We ran vertically-refined simulation cases with Brooks-Corey-like relative permeability expressions with unity endpoints relative permeabilities:

$$k_{rc} = \left(\frac{s_c}{1 - s_{wirr}} \right)^{n_c}, \quad k_{rw} = \left(\frac{s_w - s_{wirr}}{1 - s_{wirr}} \right)^{n_w} \quad (37a,b)$$

where the irreducible water saturation ($s_{wirr} = 0.3$). Once the simulation is completed, we numerically averaged the saturation and the relative permeabilities and contrasted them against the rock (fine-scale) and the analytical sharp-interface vertically-averaged relative permeabilities. As heterogeneity does not pose a first-order impact, we modelled homogeneous formations. Thus, the vertically-averaged relative permeabilities are straight lines. Also, as it is clear that the major difference between the vertically-refined and the VE models occurs at the stop of injection, we show the results for this timestep. For all of the models, there is a single VE model, its reconstructed saturation distribution is shown in Fig. 14. Fig. 15 shows the CO₂ saturation distributions, and their numerically-averaged relative permeabilities. VE saturation distribution conserves most of the features of all cases presented and, in some sense, it is the average of all.

Clearly, in all cases, the calculated vertically-averaged relative permeabilities from the vertically-refined models are nearly perfect straight-lines. However, notice the shifting happened to the vertically-averaged relative permeabilities endpoints in the cases where we varied (n_w) and in the cases where ($n_w = n_c$). Similar observation reported by Becker et al. (2017) for which a pseudo-vertical equilibrium model for slow gravity drainage was developed.

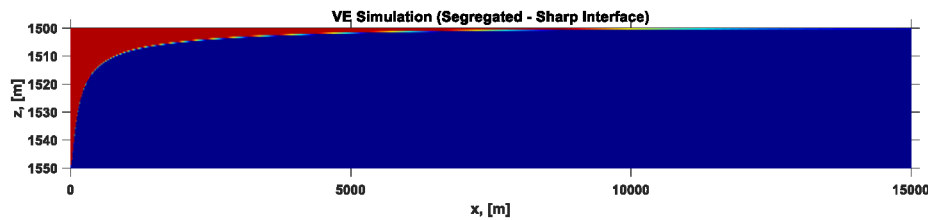


Figure 14—Reconstructed vertical equilibrium (segregated flow) CO₂ saturation.

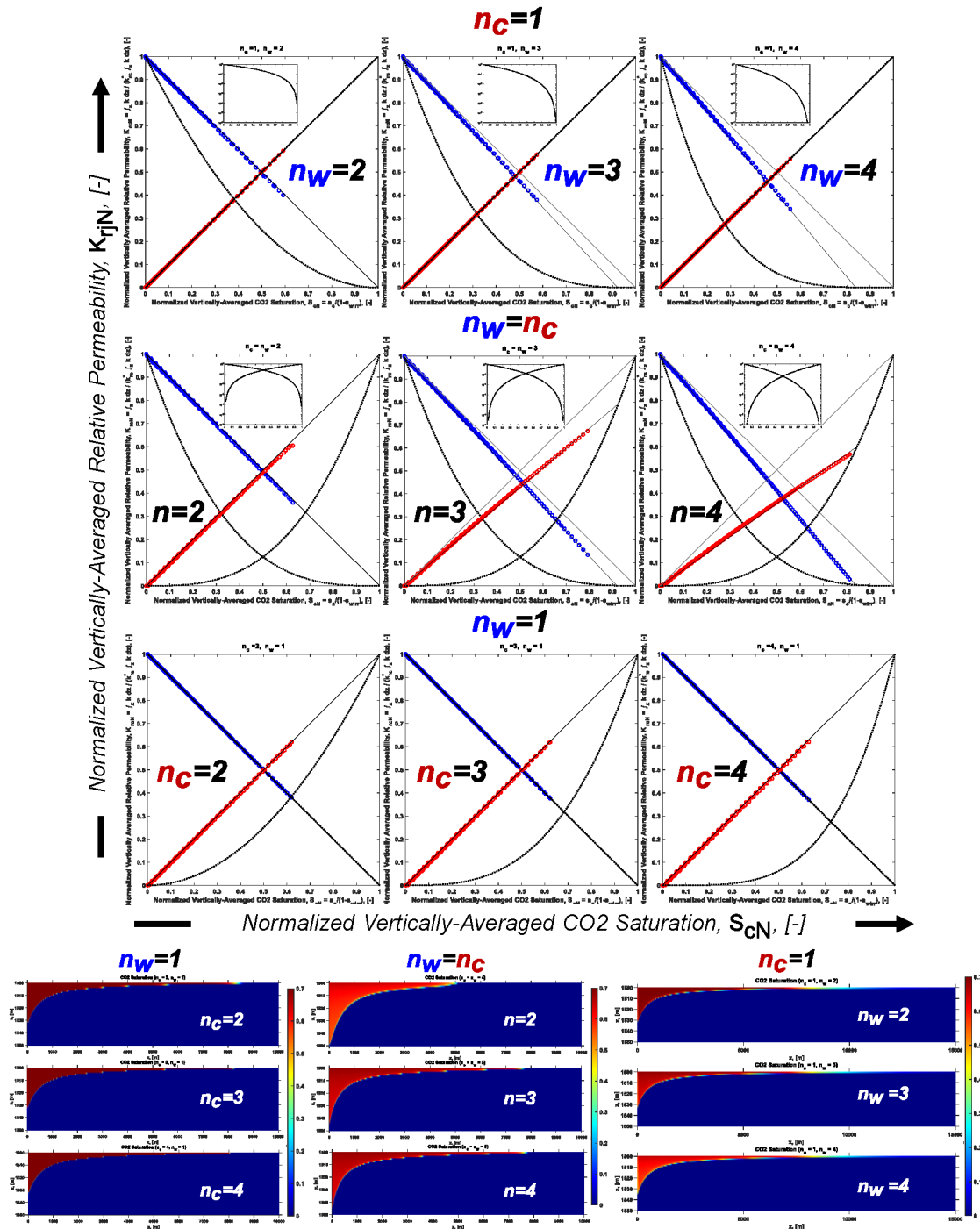


Figure 15—Vertically-averaged relative permeability (top) and CO₂ saturation distribution (bottom) of different CO₂ and water fine-scale relative permeability curvatures.

Homogeneous-Acting Formations

The present study focuses on scaling the flow equations in homogeneous media and deriving insights into the dynamic range of applicability of the different sub-models. A complete description with applications to heterogeneous formations can be found somewhere else (Alamara, Blondeau, Thibaud, et al. 2025). Still, given the averaging involved, some formations are effectively homogeneous-acting despite the underlying variations in the properties. See Fig. 16 as an example of such formations.

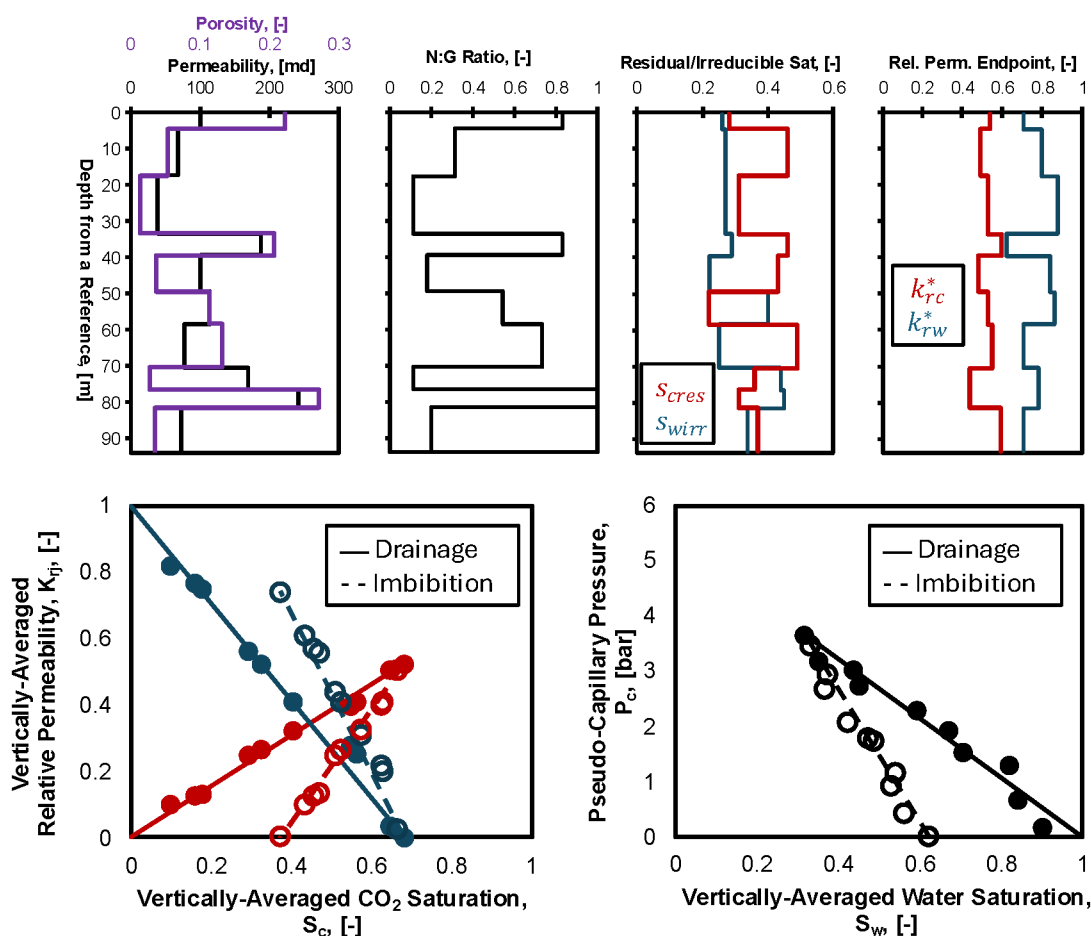


Figure 16—Petrophysical properties distributions of a synthetic model (top) and the sharp-interface vertical equilibrium flow functions (bottom). The latter were calculated using the spreadsheet provided by (Alamara, Blondeau, Thibeau, et al. 2025)

Application Cases

In this section, we present two cases where simplifying assumptions can be made. The first deals with homogeneous-acting, real field aquifer model for CO₂ storage. And the second deals with a synthetic anticline aquifer for hydrogen storage in aquifer focusing on the issue of residual trapping and grid sensitivity.

CO₂ Storage Simulation in a Homogeneous-Acting Aquifer for Screening Phase

The North-West aquifer is a projected site to store a maximum of 42 MTPA of CO₂ through 14 fully-perforated injection wells. Geometrically, the formation spans an area of 56 km x 65 km with an average thickness of 680 m and is sloping northwestward at an angle of around 1 degree regionally, see Fig 17. Geologically, the model consists of three main facies and seven geological layers whose petrophysical properties are shown in Figs. 17 and 18. The permeability is lognormally distributed while the porosity is normally distributed. The vertically aggrading pattern of the petrophysical properties allows treating the formation as homogeneous-acting during the screening phase through straight-line VE flow functions similar to those shown previously in Fig. 16. As expected at the screening phase, the 3D numerical model is coarse with average gridblock dimensions of $\Delta x = \Delta y = 280$ m and $\Delta z = 5.7$ m. The properties were averaged vertically for the VE model, see Fig. 18, and ran using the same simulator used for the 3D case - INTERSECT simulator (SLB 2024b). It is worth noting that it is possible to account for petrophysical heterogeneity through the vertically-averaged relative permeability and pseudo-capillary pressure but our focus here is to show an example where simplified assumptions can be made.

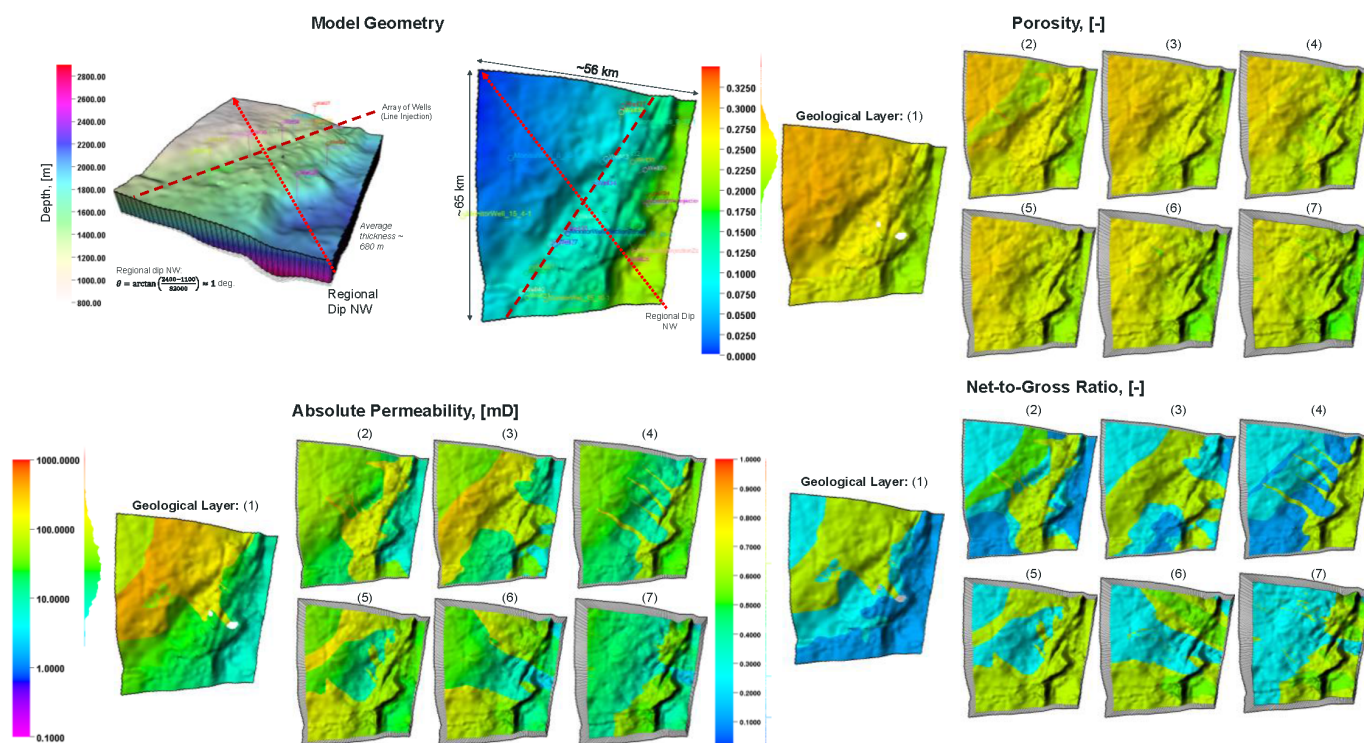


Figure 17—The North-West model's geometry (to p-left) and its geological layers' porosity (to p-right), absolute permeability (bottom-left), and net-to-gross ratio (bottom-right) distributions.

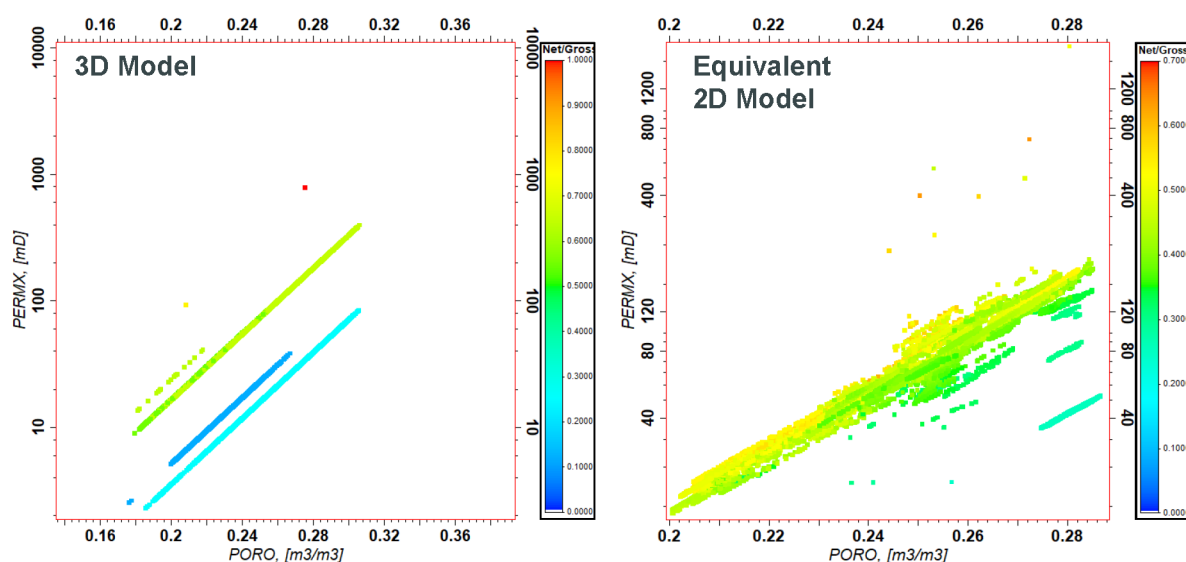


Figure 18—Permeability-porosity-NTG (net-to-gross) cross plots of the 3D models (left) and the equivalent 2D model (right).

We ran different cases of both 3D and VE models considering different physics but opted for brevity to show results of two extreme cases: (1) considering structural trapping only, and (2) considering structural, residual, and contact dissolution trapping in the 3D model and structural and residual trapping in the VE model. Fig. 19 shows the vertically-averaged CO₂ saturation of both models after 1000 years from stopping the injection. The results demonstrate qualitatively reasonable agreement for early stages of development given that further extended plume is expected from the VE model.

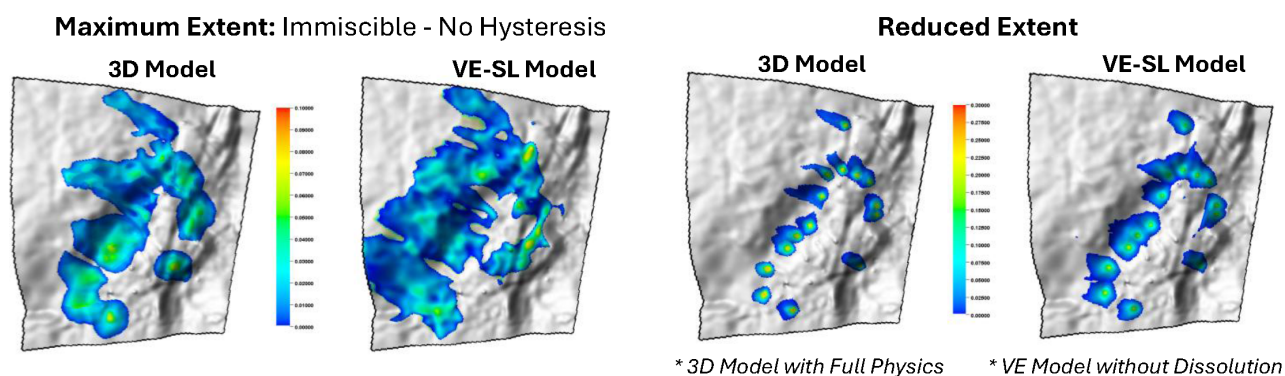


Figure 19—CO₂ saturation distributions of the 3D and the VE models after 1000 years of post-injection.

Hydrogen Storage in a Synthetic Anticline Aquifer

The use of VE for simulating hydrogen storage in aquifers has been discussed by [Becker et al. \(2022\)](#), and its applicability by [Vahabzadeh et al. \(2025\)](#). In terms of simulating the basic hydrodynamics of the process, hydrogen storage is seasonal by its nature and therefore the hydrodynamics of the injection-storage-production cycles are mainly in the near-field (aka near the wellbore) unlike CO₂ storage where the focus is mainly for long-term migration and trapping of the plume. The density difference between the phases is larger in the case of hydrogen storage as compared to the case of CO₂ storage but the capillary pressure is higher as well. We simulate a simple hydrogen injection-storage-production in a homogeneous anticline using black-oil like module of MRST simulator, consult the work of [Ahmed et al. \(2024\)](#) for a detailed treatment. [Fig. 20](#) shows the hydrogen saturation distribution at the end of injection of two different vertical grid resolutions along the reconstructed VE model at the end of the 10th cycle. The grid resolution effect is clear and highly impacts the amount of residually trapped hydrogen. The VE model provides results with theoretically infinite vertical resolution at lower computational costs and faster runtimes, hence as the 3D model's grid is refined, the results approach that of the VE model. We emphasize that care must be taken when predicting residual trapping given the level it can be overestimated in coarse 3D models. [Fig. 21](#) compares the well's rate and bottom-hole pressure where reasonable agreement can be noticed between the fine model's results and those obtained from the VE model. We must mention that we used the total gas production rate as a constraint in this example case, however, for more realistic cases a hybrid approach where the near well bore region is simulated using 3D fine-grid is generally required ([Becker et al. 2022](#)).

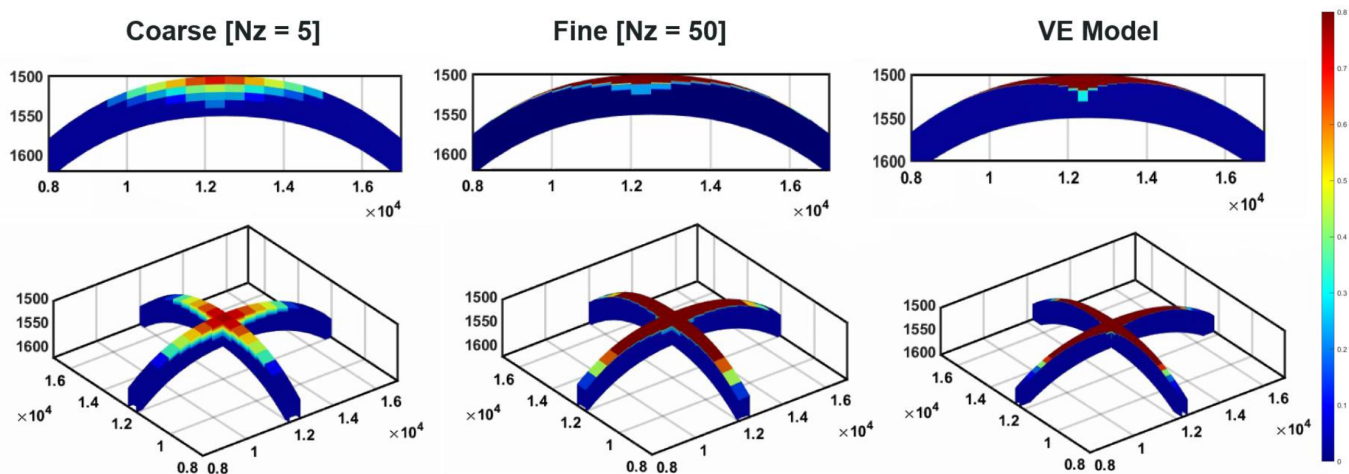


Figure 20—Impact of grid resolution on the hydrogen saturation distribution after 10th injection-storage-production cycles in a synthetic anticline. The top row shows a sliced view whilst the bottom row shows a 3D view. The coordinates' unit is meter.

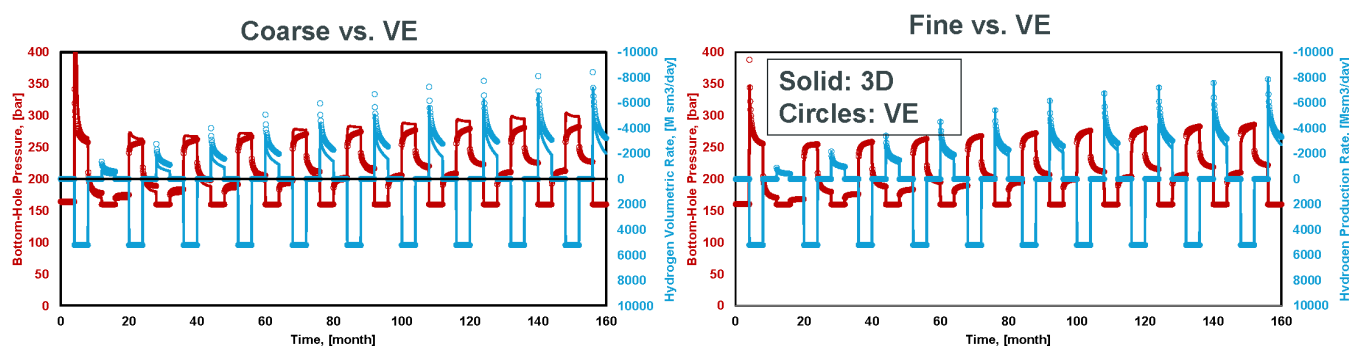


Figure 21—Well-bottomhole pressure and volumetric rate. A comparison between the coarse model vs. VE model (left) and the fine model vs. VE model (right).

Conclusions

This study combines theoretical considerations and numerical analysis to establish the operational limits and applicability of vertically-integrated models based on the vertical equilibrium (VE) assumption. We clarified the hierarchy of the underlying assumptions and explored the relationships between governing flow regimes, model formulations, and physical conditions for accurate VE application. Our findings confirm that VE is valid across the spectrum of gravity-capillarity interactions, including both sharp-interface and capillary-dispersed flows. We further distinguish between viscous-dispersed and capillary-dispersed regimes, which — despite sharing similar mathematical structures — arise from different physical mechanisms, both involving significant vertical crossflow. A critical threshold was identified for the buoyancy number $N_b > 3$, above which segregated flow and VE assumptions are justifiable. For lower N_b , we propose a generalized flow functions approach, which reasonably replicates saturation distributions from high-resolution 2D simulations. To assess the assumption of instantaneous gravity-capillarity equilibrium, we derived a mechanistic equilibrium time scale that accurately predicts the validity of VE models relative to simulation or injection durations. In the context of CO₂-brine systems, sharp-interface models may suffice for preliminary assessments, but including capillary transition zones yields more realistic predictions especially for long-term assessments. Notably, local curvature effects can change the model results. Finally, we emphasize that some formations behave as homogeneous-acting systems; the conditions under which this simplification is appropriate have been discussed in an earlier work by Alamara, Blondeau, Thibaud, et al. (2025). Application examples confirm that VE-based simplifications can offer efficient and physically consistent simulation results when applied under well-defined conditions (Alamara, Blondeau, Thibaud, et al. 2025). We concluded the study with application cases where simplifying assumption were utilized reasonably.

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